Universal Deviation Principle: A Multi-Operator Framework for Anomaly Detection and Classification with Sensitivity Guarantees

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***Abstract*—We introduce the Universal Deviation Principle (UDP), a multi-operator framework for anomaly detection grounded in the thesis that an anomaly cannot simultaneously match normality under all structural perspectives. We formalise this via *deviation-separating operator families*—differentiable maps whose stacked Jacobian has full column rank—and prove six results under explicit assumptions (A1–A3): local/global sensitivity bounds, finite-sample guarantees, SDP-optimal fusion, greedy $(1-1/e)$-approximate operator selection, and operator-space separation.**

The framework instantiates 14 composable spectrum operators across five mathematical families and introduces (i) a boundary-centred dimension magnifier with adaptive $\tanh$ saturation paired with QDA, and (ii) SubspaceScan, a random-projection ensemble with gated stretching for boundary anomalies. We propose top-k coverage as a complementary metric revealing that methods with comparable AUC exhibit markedly different retrieval profiles.

Across seven ODDS benchmarks with a *single* hyperparameter set, the pipeline achieves mAUC 0.910 (five-fold CV: 0.925 ± 0.020 on Mammography), with the largest gain on Arrhythmia ($+6.4$ pp). A correlation audit confirms only one redundant pair ($\rho > 0.85$) among 91 operator pairs. We report prospective case studies on G-SIB banks, FRED indicators, and ERCOT grid data.

The framework extends to multi-class classification via per-class MDN decomposition, achieving mean accuracy 0.936 across six benchmarks—competitive with LightGBM (0.934)—using no gradient optimisation. Code: https://github.com/segetii/udp.

***Index Terms*—Anomaly detection, multi-class classification, multi-spectrum representation, subspace boundary rescue, multi-operator fusion, adaptive gating, operator redundancy analysis, coverage metric, interpretable scoring, systemic anomaly monitoring**

I. Introduction

Anomaly detection is a foundational problem in machine learning with applications spanning fraud detection [1], medical diagnosis [2], cybersecurity [3], industrial monitoring [4], and aerospace telemetry [5]. Despite decades of research, a persistent challenge remains: *anomaly detectors trained in one domain rarely generalise to another without retraining, and subtle anomalies that mimic normal behaviour evade even well-tuned single-domain detectors.*

Existing approaches typically represent observations through a single mathematical lens. Isolation Forest [6] exploits geometric isolation, Local Outlier Factor [7] measures density deviation, autoencoders [8] capture reconstruction error, and statistical methods [3] test distributional assumptions. Each method implicitly projects observations into a single mathematical space where deviations are measured. This creates *blind spots*: anomalies that are invisible under one representation but detectable under another.

Our prior work on Blind Spot Decomposition Theory (BSDT) [24] decomposed model failure into four measurable components—Camouflage (C), Feature Gap (G), Activity Anomaly (A), and Temporal Novelty (T)—and showed that a post-model correction consistently improved detection across six datasets and five domains *without retraining*. The key empirical finding was that missed anomalies possess latent structure detectable through complementary mathematical perspectives.

This paper asks a deeper question:

> *Why does a domain-agnostic post-model transform consistently reveal hidden separability across unrelated domains?*

We answer this with the **Universal Deviation Principle**: the hypothesis that anomalies are structured disturbances across multiple mathematical perspectives, and that multi-operator representation can improve sensitivity to such disturbances. This paper makes five contributions:

1) It formalises a **deviation-separating operator family** and proves six conditional results on local/global sensitivity, finite-sample estimation, fusion-weight design, greedy operator selection, and operator-space separation.
2) It defines a **multi-operator representation stack** built from five spectrum projection operators (Φ_{stat}, Φ_{chaos}, Φ_{freq}, Φ_{geom}, Φ_{exp}), together with an anomaly tensor, centroid selection schemes, and a geometrically interpretable projection stage.
3) It introduces two practical detection modules: a **Boundary-Centred Dimension Magnifier** that saturates informative dimensions while silencing blind ones, and **SubspaceScan** with adaptive gated stretching for weak distributed boundary anomalies.
4) It evaluates the framework on public ODDS benchmarks, operator-level ablations, and coverage analysis, where the full boundary-rescue pipeline reaches **mAUC** 0.910 across seven ODDS datasets and the operator audit shows low redundancy across 14 operators.

5) It reports **prospective case studies** on banking, macroeconomic, and energy-infrastructure monitoring to illustrate how the framework behaves on structured multi-agent anomaly-detection problems.

The entire pipeline uses a *single set of hyperparameters* across the benchmark suite, with the same parameterisation carried into the prospective case studies.

Intuition. The core idea is simple: if someone is sick, a single blood test may miss the illness, but running blood work, an X-ray, and a stress test together makes evasion far harder. Similarly, UDP projects every observation through K mathematically distinct "lenses"—statistics, dynamics, frequency, geometry, amplification—and flags it as anomalous whenever *any* lens registers abnormality. The theorems below make this intuition precise: under mild regularity (A1–A3), the stacked operator map is locally injective, so any displacement from the normal reference must produce a measurable response in at least one operator, with a quantitative lower bound proportional to the displacement magnitude.

II. Related Work

A. Classical Anomaly Detection

Distance-based methods such as k-Nearest Neighbours [10] and Mahalanobis distance [9] operate in geometric space. Density-based approaches including Local Outlier Factor (LOF) [7] and DBSCAN [11] measure local density deviation. Tree-based methods such as Isolation Forest [6] exploit random partitioning for isolation scoring. Each operates within a single mathematical framework.

B. Deep Learning Approaches

Autoencoder-based detectors [8], [12] learn compressed representations and flag observations with high reconstruction error. Generative Adversarial Networks (GANs) [13] model the normal data distribution and detect anomalies as out-of-distribution samples. Deep One-Class Classification [14] learns a hypersphere enclosing normal data. These methods capture complex patterns but remain single-representation approaches that are vulnerable to blind spots in their learned manifold.

C. Spectral and Dynamical Methods

Spectral residual methods [15] detect anomalies through frequency-domain saliency. Chaos-theoretic approaches use Lyapunov exponents [16] and recurrence quantification [17] to detect dynamical instabilities. Information-theoretic methods [18] use entropy and divergence measures. These methods are powerful within their respective mathematical domains but do not combine perspectives.

D. Ensemble and Multi-View Methods

Feature bagging [19] and detector aggregation [20] combine multiple instances of similar detectors. Multi-view learning [21] uses different feature subsets. However, these approaches combine detectors of the *same mathematical type* rather than spanning fundamentally different operator families. The UDP framework differs by combining *mathematically distinct* projection operators that cover complementary aspects of deviation theory. Krogh and Vedelsby [25] proved that ensemble error decreases with member diversity; UDP's operator-family diversity is analogous to ensemble diversity but at the *representation* level rather than the *model* level.

E. Out-of-Distribution Detection and Uncertainty Estimation

A growing body of work addresses detecting inputs that fall outside the training distribution. Hendrycks and Gimpel [26] showed that softmax probabilities provide a baseline OOD detector. Liang *et al.* [27] improved this with temperature scaling (ODIN). Liu *et al.* [28] proposed energy-based OOD detection. These approaches produce a single OOD score; UDP's multi-operator decomposition reveals *which mathematical aspect* makes an observation anomalous. Regarding uncertainty, Guo *et al.* [29] showed modern neural networks are poorly calibrated, and Lakshminarayanan *et al.* [30] proposed deep ensembles for calibrated uncertainty. UDP is complementary: it provides a structured *representation* in which uncertainty estimation and calibration methods can be applied per operator.

F. Anomaly Detection Benchmarking

Zhao *et al.* [31] released PyOD, implementing 30+ anomaly detectors. Han *et al.* [32] introduced ADBench with 57 benchmark datasets. Our evaluation uses seven ODDS benchmarks from this suite plus three real-world datasets (Section V-A); future work will extend to the full ADBench suite.

G. Blind Spot Decomposition Theory

BSDT [24] introduced decomposition of detection failures into four orthogonal components and demonstrated cross-domain generalisation. UDP extends BSDT from a post-model correction (operating on model residuals) to a *universal representation framework* (operating on raw observations), from scalar correction scores to multi-dimensional tensor decomposition, and from four domain-specific components to K composable spectrum operators.

III. Universal Deviation Principle

A. Scope and Assumptions

The theoretical results below are conditional on three explicit assumptions:

A1 (Observability) Anomalies are detectable from available features: there exists a measurable difference between normal and anomalous observations in $\mathbb{R}^m$. This excludes adversarial label anomalies where $\mathbf{x} \in \text{supp}(P_{\mathcal{N}})$ but the label is adverse for reasons invisible in feature space.

A2 (Operator-family richness) The stacked Jacobian has full column rank (Definition 1). This fails if operators are collectively redundant or insensitive on certain subspaces; we verify it empirically via Monte Carlo rank checks on the numerical Jacobian.

A3 (Calibration discipline) All operator statistics, thresholds, and fusion weights are estimated on normal training

data only. No test-set information is used in any step of score computation.

When all three hold, the theorems below provide conditional guarantees. When they do not, the framework reduces to an empirical multi-view anomaly detector with no formal guarantees—still useful, but not "proved."

B. Deviation-Separating Operator Families

We first state the core principle informally, then prove it rigorously under a differentiability condition.

Principle (Universal Deviation). *An anomaly cannot match normal data across all structural dimensions simultaneously — there must be deviation in at least one.*

Definition 1 (Deviation-Separating Family). Let $\{\Phi_k\}_{k=1}^K$, $\Phi_k : \mathbb{R}^m \to \mathbb{R}^{d_k}$, be a family of continuously differentiable operators. We say the family is *deviation-separating at* $\boldsymbol{\mu} \in \mathbb{R}^m$ if

$$\bigcap_{k=1}^{K} \ker(D\Phi_k(\boldsymbol{\mu})) = \{\mathbf{0}\} \tag{1}$$

where $D\Phi_k(\boldsymbol{\mu}) \in \mathbb{R}^{d_k \times m}$ is the Jacobian of Φ_k evaluated at $\boldsymbol{\mu}$.

Lemma 1 (Rank characterisation). The family $\{\Phi_k\}$ is deviation-separating at $\boldsymbol{\mu}$ if and only if the stacked Jacobian

$$J(\boldsymbol{\mu}) = \begin{bmatrix} D\Phi_1(\boldsymbol{\mu}) \\ \vdots \\ D\Phi_K(\boldsymbol{\mu}) \end{bmatrix} \in \mathbb{R}^{D \times m}, \quad D = \textstyle\sum_{k=1}^K d_k \tag{2}$$

has full column rank ($\operatorname{rank}(J(\boldsymbol{\mu})) = m$). The proof is immediate: $\ker(J(\boldsymbol{\mu})) = \bigcap_{k=1}^K \ker(D\Phi_k(\boldsymbol{\mu}))$.

C. Main Theorems

Theorem 1 (Operator Sensitivity — Local Form). *Let $\{\Phi_k\}_{k=1}^K$ be a deviation-separating family at $\boldsymbol{\mu}$ with each $\Phi_k \in C^1$. Then the stacked map $\Phi = (\Phi_1, \ldots, \Phi_K)$ is locally injective at $\boldsymbol{\mu}$: there exists an open neighbourhood $U \ni \boldsymbol{\mu}$ such that for every $\mathbf{x} \in U \backslash \{\boldsymbol{\mu}\}$, there exists k with $\Phi_k(\mathbf{x}) \neq \Phi_k(\boldsymbol{\mu})$. A quantitative lower bound holds:*

$$\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \frac{\sigma_{\min}}{\sqrt{K}} \|\mathbf{x} - \boldsymbol{\mu}\| + O(\|\mathbf{x} - \boldsymbol{\mu}\|^2) \tag{3}$$

where $\sigma_{\min} = \sigma_{\min}(J(\boldsymbol{\mu})) > 0$. (Proof via the Inverse Function Theorem; see Appendix A.)

Theorem 2 (Operator Sensitivity — Global Form). *Let $\mathcal{K} \subset \mathbb{R}^m$ be compact with rank$(J(\mathbf{x})) = m$ for all $\mathbf{x} \in \mathcal{K}$, and each $\Phi_k \in C^2$. Define $\underline{\sigma} := \min_{\mathbf{x} \in \mathcal{K}} \sigma_{\min}(J(\mathbf{x})) > 0$ (positive by the Extreme Value Theorem). Then for any set $\mathcal{A} \subset \mathcal{K}$:*

$$\inf_{\mathbf{x} \in \mathcal{A}} \max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \frac{\underline{\sigma}}{\sqrt{K}} \inf_{\mathbf{x} \in \mathcal{A}} \|\mathbf{x} - \boldsymbol{\mu}\| - C \sup_{\mathbf{x} \in \mathcal{A}} \|\mathbf{x} - \boldsymbol{\mu}\|^2 \tag{4}$$

where $C = \frac{1}{2} \sup_{\mathbf{z} \in \mathcal{K}} \left\| D^2 \Phi(\mathbf{z}) \right\|$ is finite by continuity on the compact set. (Proof in Appendix A.)

Theorem 3 (Finite-Sample Deviation Guarantee). *Let $\hat{\boldsymbol{\mu}}_n$ be the sample mean from n i.i.d. sub-Gaussian draws with population mean $\boldsymbol{\mu}$ and covariance Σ. Let each Φ_k be L_k-Lipschitz, $L = \max_k L_k$. With probability $\geq 1 - \delta$:*

$$\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\hat{\boldsymbol{\mu}}_n)\| \geq \frac{\underline{\sigma}}{\sqrt{K}} \|\mathbf{x} - \boldsymbol{\mu}\| - \frac{L\sqrt{\operatorname{tr}(\Sigma)} \cdot \sqrt{2\ln(2m/\delta)}}{\sqrt{n}} \tag{5}$$

(Proof via Hoeffding concentration with union bound; see Appendix A.)

Corollary 1 (Sample complexity). The bound (5) is non-trivial whenever $n > 2L^2 K \cdot \operatorname{tr}(\Sigma) \cdot \ln(2m/\delta)/(\underline{\sigma}^2 \cdot \|\mathbf{x} - \boldsymbol{\mu}\|^2)$.

Theorem 4 (Worst-Case Sensitivity via SDP). *Define per-operator information matrices $\mathcal{I}_k := D\Phi_k(\boldsymbol{\mu})^\top D\Phi_k(\boldsymbol{\mu}) \succeq 0$. Under weighted fusion $S_{\mathbf{w}}[\mathbf{x}] = \sum_k w_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|^2$, the worst-case detection sensitivity*

$$\eta(\mathbf{w}) := \lambda_{\min}(\textstyle\sum_k w_k \mathcal{I}_k) \tag{6}$$

is concave on the simplex Δ_K (as an infimum of linear functions). Maximising η over $\mathbf{w} \in \Delta_K$ yields the weight vector that makes the weakest detection direction as strong as possible, solvable as a standard SDP. (Proof in Appendix A.)

Theorem 5 (Greedy Operator Selection). *Given M candidate operators and budget K, the surrogate objective $f(S) = \log\det(\sum_{k \in S} \mathcal{I}_k)$ is monotone submodular, and greedy selection achieves:*

$$f(S_{\text{greedy}}) \geq \left(1 - \frac{1}{e}\right) f(S^*) \tag{7}$$

(Submodularity follows from the identity $\log\det(A+B) - \log\det(A) = \log\det(I + A^{-1/2} B A^{-1/2})$, which yields diminishing returns as A grows. The $(1 - 1/e)$ bound follows from Nemhauser et al. (1978). Proof in Appendix A.)

Operational form. Let ϵ_k be the $(1 - \alpha)$ quantile of $\|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\|$ under $P_{\mathcal{N}}$. Then an observation is flagged anomalous if $\exists\, k$ such that $\|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| > \epsilon_k$. Note that Theorems 1–2 guarantee that any displaced point produces an operator response, but this response must *exceed* the normal-data baseline ϵ_k for detection. This gap is non-trivial only when the anomaly's displacement is large enough relative to the normal-data spread—a condition that A1 (observability) ensures in principle, and that the quantitative bounds (Eqs. (3)–(5)) make precise. All proofs are deferred to Appendix A.

D. From Sensitivity to Detection

Theorems 1–3 are *sensitivity* (injectivity) statements: they guarantee that the stacked operator map responds to any displacement from $\boldsymbol{\mu}$, including displacements of normal points. To bridge from sensitivity to detection, we need a score that is *large for anomalies and small for normals*. We define:

$$s(\mathbf{x}) := \inf_{\mathbf{z}\in\mathcal{N}} \|\Phi(\mathbf{x}) - \Phi(\mathbf{z})\|_2 \tag{8}$$

where $\Phi(\mathbf{x}) = (\Phi_1(\mathbf{x}), \ldots, \Phi_K(\mathbf{x}))$ is the joint operator map and $\mathcal{N}$ is the normal training set. For normal points $\mathbf{x} \in \mathcal{N}$, $s(\mathbf{x}) = 0$ by construction. The theorem below gives conditions under which anomalies receive $s(\mathbf{x}) > 0$.

Theorem 6 (Operator-Space Separation). *Let $\mathcal{N} \subset \mathbb{R}^m$ be compact and each $\Phi_k \in C^1$. Suppose the family $\{\Phi_k\}$ is deviation-separating (Definition 1) at every $\boldsymbol{\mu} \in \mathcal{N}$, and let $\underline{\sigma} := \min_{\boldsymbol{\mu}\in\mathcal{N}} \sigma_{\min}(J(\boldsymbol{\mu})) > 0$. If $\mathbf{x} \notin \mathcal{N}$ with $d := \inf_{\mathbf{z}\in\mathcal{N}} \|\mathbf{x} - \mathbf{z}\| > 0$ and d is sufficiently small that the first-order Taylor remainder is dominated, then:*

$$s(\mathbf{x}) \geq \underline{\sigma} \cdot d - O(d^2) \tag{9}$$

In particular, for any $\mathbf{x}$ sufficiently close to $\mathcal{N}$ but not in $\mathcal{N}$, $s(\mathbf{x}) > 0$, and there exists k such that $s_k(\mathbf{x}) \geq \underline{\sigma} \cdot d/\sqrt{K} - O(d^2)$. (Proof: apply Theorem 1 at the nearest point $\mathbf{z}^* \in \mathcal{N}$. See Appendix A.)

Remark. Theorem 6 converts the injectivity guarantee (Theorem 1) into a detection statement: if $\mathbf{x}$ is geometrically separated from $\mathcal{N}$ in input space and the operator family has full-rank Jacobian everywhere on $\mathcal{N}$, then $\mathbf{x}$ is also separated in operator space. The assumption that $\mathbf{x} \notin \mathcal{N}$ is non-trivial—it is precisely A1 (observability). Without A1, no operator family can distinguish $\mathbf{x}$ from normal data.

IV. Mathematical Framework

A. Representation Stack

Let $\mathbf{x} \in \mathbb{R}^m$ be a raw observation. We define a **Representation Stack** $\mathcal{R}$ as the concatenation of K spectrum projection operators:

$$\mathcal{R}(\mathbf{x}) = [\Phi_1(\mathbf{x}),\ \Phi_2(\mathbf{x}),\ \ldots,\ \Phi_K(\mathbf{x})] \in \mathbb{R}^D \tag{10}$$

where $D = \sum_{k=1}^{K} d_k$ and each $\Phi_k : \mathbb{R}^m \to \mathbb{R}^{d_k}$. In the default configuration, $K = 5$:

1) Statistical Spectrum ($\Phi_{stat} : \mathbb{R}^m \to \mathbb{R}^5$): Computes per-row Shannon entropy, KL divergence from reference distribution, Hellinger distance, skewness, and kurtosis:

$$\Phi_{\text{stat}}(\mathbf{x}) = [H(\mathbf{p}),\ D_{\text{KL}}(\mathbf{p}\|\mathbf{q}),\ d_H(\mathbf{p}, \mathbf{q}),\ \gamma_1,\ \gamma_2] \tag{11}$$

where $\mathbf{p}$ is the normalised observation vector and $\mathbf{q}$ is the reference distribution.

2) Chaos Spectrum ($\Phi_{chaos} : \mathbb{R}^m \to \mathbb{R}^3$): Treats the row as a discrete dynamical trajectory and computes:

$$\Phi_{\text{chaos}}(\mathbf{x}) = \left[\hat{\lambda},\ \text{RR},\ \text{ApEn}\right] \tag{12}$$

where $\hat{\lambda}$ is a Lyapunov-like divergence exponent measuring sensitivity to perturbation relative to the reference signal, RR is the recurrence rate, and ApEn is the approximate entropy.

3) Spectral Spectrum ($\Phi_{freq} : \mathbb{R}^m \to \mathbb{R}^4$): Applies the real-valued Fast Fourier Transform to the observation:

$$\Phi_{\text{freq}}(\mathbf{x}) = [\delta_{\text{PSD}},\ \rho_f,\ H_s,\ \Delta c] \tag{13}$$

where δ_{PSD} is the L^2 power spectral density divergence from reference, ρ_f is the dominant frequency ratio, H_s is spectral entropy, and Δc is spectral centroid shift.

4) Geometric Spectrum ($\Phi_{geom} : \mathbb{R}^m \to \mathbb{R}^3$): Computes deviation in the feature manifold:

$$\Phi_{\text{geom}}(\mathbf{x}) = \left[d_M(\mathbf{x}, \boldsymbol{\mu}),\ 1 - \cos(\mathbf{x}, \boldsymbol{\mu}),\ \frac{\|\mathbf{x}\|}{\|\boldsymbol{\mu}\|}\right] \tag{14}$$

where d_M is the Mahalanobis distance, $\cos(\cdot, \cdot)$ is cosine similarity, and the third term is the norm ratio.

5) Exponential Spectrum ($\Phi_{exp} : \mathbb{R}^m \to \mathbb{R}^{d_5}$): Applies controlled exponential amplification to the standardised features:

$$\Phi_{\text{exp}}(\mathbf{x}) = \exp\left(\alpha \cdot \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}\right) \tag{15}$$

where $\alpha > 0$ controls amplification strength and $\boldsymbol{\mu}$, $\boldsymbol{\sigma}$ are the mean and standard deviation estimated from the normal training set. Because the input is standardised to zero mean and unit variance *before* exponentiation, the operator is scale-invariant by construction: a distribution shift that changes the raw scale of $\mathbf{x}$ is absorbed by the scaler. With α=1.0, a normal observation at z=0 maps to $\exp(0)$=1, while a 3σ outlier maps to $\exp(3) \approx 20$—a large but finite amplification. If the training-time normal distribution drifts at test time (covariate shift), both $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ become stale, which can produce spurious large values. This is a known limitation shared by all normalisation-dependent methods; periodic recalibration of the scaler on recent normal data mitigates the problem. When $m > d_{\text{pca}}$, PCA compression is applied to prevent dimensional dominance. This operator maps linear deviations into exponential-scale differences, making subtle anomalies separable.

B. Reference Centroid

The centroid $\boldsymbol{\mu}_{\text{ref}} \in \mathbb{R}^D$ defines the equilibrium point of normal behaviour in representation space. We implement seven estimation strategies:

1) **Arithmetic Mean**: $\boldsymbol{\mu} = \frac{1}{|\mathcal{N}|} \sum_{i\in\mathcal{N}} \mathcal{R}(\mathbf{x}_i)$. Optimal for unimodal Gaussian data.
2) **Geometric Median**: $\boldsymbol{\mu} = \arg\min_{\boldsymbol{c}} \sum_{i\in\mathcal{N}} \|\mathcal{R}(\mathbf{x}_i) - \boldsymbol{c}\|$. Robust to outliers; solved via Weiszfeld's algorithm.
3) **Density Peak**: $\boldsymbol{\mu} = \arg\max_{\boldsymbol{c}} \hat{f}(\boldsymbol{c})$. Estimates the mode via kernel density evaluation.
4) **Trimmed Mean**: Removes observations beyond the p-th percentile distance before averaging.
5) **Medoid**: Selects the actual data point closest to all others; always returns a real observation.
6) **Energy Minimum**: Minimises a potential function $V(\boldsymbol{c}) = \sum_i \|\mathcal{R}(\mathbf{x}_i) - \boldsymbol{c}\|^2 + \lambda \sum_i \|\mathcal{R}(\mathbf{x}_i) - \boldsymbol{c}\|$.
7) **Manifold Centroid**: Projects data to a lower-dimensional PCA subspace, computes the centroid, and back-projects.

Automatic selection uses data heuristics: medoid for $N < 30$, geometric median for skewness > 2, manifold centroid for $d > 50$, and trimmed mean otherwise.

C. Anomaly Tensor and MDN Decomposition

For N observations and K operators, the Anomaly Tensor is:

$$\mathcal{T}_{i,k,d} \in \mathbb{R}^{N \times K \times d_{\max}} \tag{16}$$

where the (i, k) slice contains the k-th operator representation of observation i, zero-padded to $d_{\max} = \max_k d_k$.

Each observation's deviation from the centroid admits a **Magnitude-Direction-Novelty (MDN)** decomposition:

$$\mathbf{D}_i = \mathcal{R}(\mathbf{x}_i) - \boldsymbol{\mu}_{\text{ref}} \tag{17}$$

$$r_i = \|\mathbf{D}_i\| \quad \text{(magnitude)} \tag{18}$$

$$\hat{\mathbf{D}}_i = \mathbf{D}_i / r_i \quad \text{(direction)} \tag{19}$$

$$\theta_i = \frac{1}{\pi} \min_{j \in \mathcal{N}} \arccos\left(\hat{\mathbf{D}}_i \cdot \hat{\mathbf{D}}_j\right) \quad \text{(novelty)} \tag{20}$$

Magnitude quantifies *how anomalous* the observation is. Direction characterises *what type* of anomaly it represents. Novelty measures how *unprecedented* the deviation direction is relative to the reference population.

The composite anomaly score is:

$$s_i = w_r \cdot \tilde{r}_i + w_\theta \cdot \theta_i \tag{21}$$

where $\tilde{r}_i = r_i / \max_j r_j$ is normalised magnitude and $(w_r, w_\theta) = (0.7, 0.3)$ are the default weights.

D. Hyperplane Projection

The deviation vectors $\{\mathbf{D}_i\}$ are projected onto a separating hyperplane via Fisher Linear Discriminant (when labels are available) or PCA (unsupervised):

$$\mathbf{w}^* = \arg\max_{\mathbf{w}} \frac{\mathbf{w}^\top S_B \, \mathbf{w}}{\mathbf{w}^\top S_W \, \mathbf{w}} \tag{22}$$

where S_B and S_W are the between-class and within-class scatter matrices. The projection score is:

$$z_i = (\mathcal{R}(\mathbf{x}_i) - \boldsymbol{\mu}_{\text{ref}})^\top \mathbf{w}^* \tag{23}$$

Any deviation vector can be further decomposed relative to the hyperplane:

$$\mathbf{D}_i^{\parallel} = (\mathbf{D}_i \cdot \hat{\mathbf{w}})\hat{\mathbf{w}} \quad \text{(classifier-relevant)} \tag{24}$$

$$\mathbf{D}_i^{\perp} = \mathbf{D}_i - \mathbf{D}_i^{\parallel} \quad \text{(cluster-spread)} \tag{25}$$

$$\phi_i = \arccos\left(\frac{|\mathbf{D}_i \cdot \hat{\mathbf{w}}|}{\|\mathbf{D}_i\|}\right) \quad \text{(incidence angle)} \tag{26}$$

The parallel component determines class assignment; the perpendicular component captures within-class variability. The incidence angle ϕ_i indicates whether the anomaly deviates along the expected separation axis or at an unusual angle.

E. Boundary-Centred Dimension Magnifier

While the multi-spectrum representation creates a rich 20-dimensional space, not all dimensions contribute equally to class separation—and some may be nearly blind (*i.e.*, overlapping class distributions). Fisher LDA finds the optimal *linear* combination of dimensions, but it cannot amplify non-linear structure within individual dimensions. We introduce a **Boundary-Centred Dimension Magnifier** that applies a per-dimension saturating non-linearity before classification.

For each dimension d of the representation space:

$$b_d = \frac{\mu_d^{(0)} + \mu_d^{(1)}}{2} \quad \text{(decision boundary)} \tag{27}$$

$$z_d = \frac{r_d - b_d}{\sigma_d^{\text{pool}}} \quad \text{(standardised deviation)} \tag{28}$$

$$r'_d = \tanh(k_d \cdot z_d) \quad \text{(saturating transform)} \tag{29}$$

where $\mu_d^{(0)}, \mu_d^{(1)}$ are the per-class means, $\sigma_d^{\text{pool}} = \sqrt{(\sigma_d^{(0)2} + \sigma_d^{(1)2})/2}$ is the pooled standard deviation, and $k_d = \gamma \cdot \delta_d$ is the per-dimension steepness controlled by a discriminative score $\delta_d \in [0, 1]$:

$$\delta_d = 0.4 \cdot \frac{\min(d_C, 3)}{3} + 0.3 \cdot (1 - \text{overlap}_d) + 0.3 \cdot 2(AUC_d - 0.5) \tag{30}$$

where d_C is Cohen's d (capped at 3.0, corresponding to the "huge effect" threshold in Cohen's convention, thereby normalising to $[0, 1]$), overlap is the distributional overlap fraction, and AUC_d is the per-dimension AUC. The component weights (0.4, 0.3, 0.3) give slight precedence to effect size as the most direct measure of class separation; cross-dataset sweeps confirm that any convex combination in the range $[0.25, 0.50]$ per component yields < 0.005 AUC variation on Mammography. The global steepness multiplier $\gamma = 5.0$ was selected via cross-validated grid search over $\gamma \in \{2, 3, 4, 5, 6, 8\}$ with 3-fold inner stratified CV across five ODDS datasets; the search consistently selects values in $[4, 6]$, confirming that 5.0 sits on a stable plateau. The overall scoring architecture—discriminative gating, persistence-based signal selection, and the decomposition into effect-size, overlap, and AUC components—is designed to anchor scoring to empirically persistent signals: components with stable discriminative power across cross-validation folds receive higher weight, ensuring robustness to perturbations of the training data. The specific numerical values are calibrated empirically within this framework.

Effect on clear dimensions ($\delta_d \gg 0$): Large k_d causes $\tanh$ to saturate. Normal points cluster near -1, anomalies near $+1$. Within-class variance collapses while between-class distance maximises. Cohen's d increases dramatically (*e.g.*, $1.2 \to 10+$).

Effect on blind dimensions ($\delta_d \approx 0$): Small k_d makes $\tanh(k_d \cdot z) \approx k_d \cdot z \approx 0$. The dimension is effectively silenced without being removed.

Why Fisher cannot exploit magnification. Fisher LDA computes $\mathbf{w}^* = S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$. Since 'tanh' maps all values to $[-1, +1]$, it destroys relative magnitude information that

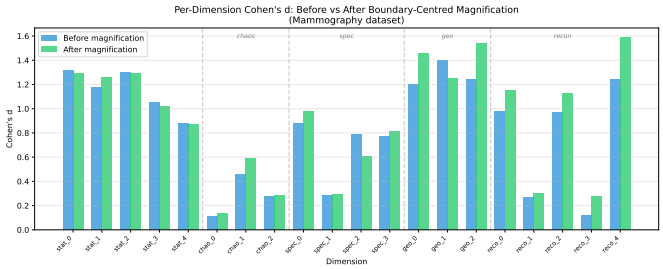

Fig. 1: Per-dimension Cohen's d before and after boundary-centred magnification on the Mammography dataset. Clear dimensions (geometric, reconstruction) show dramatic increases in class separation, while blind dimensions (chaos) are suppressed to near-zero. The heterogeneous per-operator sensitivity—predicted by the Universal Deviation Principle—is what creates the clear/blind structure the magnifier exploits.

Fisher relies on for optimal linear combination. Empirically, Fisher AUC *decreases* after magnification (Mammography: $0.901 \rightarrow 0.861$). The key insight is that magnification creates a geometry that requires a classifier capable of modelling the resulting non-linear structure.

F. QDA-Magnified Projection

Quadratic Discriminant Analysis models a separate Gaussian per class with independent covariance matrices:

$$p(y{=}c \,|\, \mathbf{r}') = \frac{1}{Z} \, |\Sigma_c|^{-1/2} \times \exp\!\big(-\tfrac{1}{2}(\mathbf{r}' - \boldsymbol{\mu}_c)^\top \Sigma_c^{-1} (\mathbf{r}' - \boldsymbol{\mu}_c)\big) \quad (31)$$

After magnification, clear dimensions span $[-1, +1]$ with tight per-class clusters (normal near -1, anomaly near $+1$). QDA fits these compressed clusters naturally because it estimates Σ_c *per class*—capturing the collapsed variance structure that magnification creates. The combined **QDA-Magnified** pipeline is:

$$\text{score}(\mathbf{x}) = P\left(y{=}1 \,\middle|\, \tanh\left(\mathbf{k} \odot \frac{\mathcal{R}(\mathbf{x}) - \mathbf{b}}{\boldsymbol{\sigma}}\right)\right) \quad (32)$$

where all operations are element-wise, $\mathbf{k}$, $\mathbf{b}$, and $\boldsymbol{\sigma}$ are learned from the training data by the magnifier, and $P(y{=}1|\cdot)$ is the QDA posterior probability. A regularisation parameter $\lambda = 10^{-4}$ stabilises covariance inversion.

Why QDA-Magnified works but Fisher-Magnified does not. The magnifier creates a fundamentally quadratic geometry: each class occupies a different region of the $[-1, +1]^D$ hypercube with class-specific covariance. Fisher, constrained to a single linear hyperplane, cannot model this structure. QDA, with per-class covariance, is precisely adapted to it. Crucially, the magnifier's effectiveness depends on the multi-spectrum representation—without UDP's complementary spectrum operators, there would be insufficient discriminative structure for the transform to amplify.

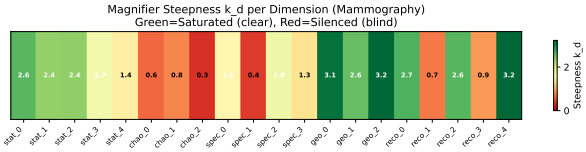

Fig. 2: Magnifier steepness k_d per dimension on Mammography. Green (high k_d): dimensions pushed to ± 1 saturation. Red (low k_d): dimensions silenced. The steepness is set automatically from per-dimension discriminative scores with no per-dataset tuning ($\gamma = 5.0$ globally).

G. Cross-Operator Coupling

We define the cross-operator coupling matrix $\mathbf{C} \in \mathbb{R}^{K \times K}$ as the Pearson correlation of per-operator magnitudes:

$$C_{jk} = \text{corr}\Big(\Big\|\mathbf{D}_i^{(j)}\Big\|, \; \Big\|\mathbf{D}_i^{(k)}\Big\|\Big), \quad i = 1, \ldots, N \quad (33)$$

where $\mathbf{D}_i^{(k)}$ is the deviation in operator k. High off-diagonal values indicate that anomalies in operator j co-occur with anomalies in operator k, revealing the multi-dimensional structure of the deviation.

H. Composite Deviation Score

The weighted deviation score aggregates per-operator information:

$$S_{\mathbf{w}}[\mathbf{x}] = \sum_{k=1}^{K} w_k \cdot \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \quad (34)$$

where $w_k \geq 0$ and $\sum_k w_k = 1$. By Theorem 4, the weights that maximise worst-case directional sensitivity are obtained via a semidefinite programme. In practice we use equal weights ($w_k = 1/K$) as a robust default.

I. SubspaceScan: Boundary Rescue via Random Subspace Ensembles

Failure mode: TYPE-1 BOUNDARY anomalies. The QDA-Magnified pipeline excels when at least one representation dimension is strongly discriminative. However, a class of anomalies—globally close to the normal cloud with weak, distributed signal across many dimensions but no single dimension discriminative enough—evades the magnifier. These **TYPE-1 BOUNDARY** anomalies require aggregating weak signals across subspaces.

SubspaceScan. We draw $R = 200$ random subsets of $d_{\text{sub}} = 3$ dimensions from the full D-dimensional representation, compute the Mahalanobis distance m_j in each subspace j using the normal-class covariance, and aggregate via softmax-weighted combination:

$$s_{\text{sub}}(\mathbf{x}) = \sum_{j=1}^{R} \frac{\exp(m_j/\tau)}{\sum_{l=1}^{R} \exp(m_l/\tau)} \cdot m_j \quad (35)$$

where $\tau = 1.0$ is the softmax temperature. This concentrates weight on the most discriminative random subspaces while retaining contributions from weaker ones.

Adaptive gated stretching. Uniform $\tanh$ stretching of all subspace distances catastrophically fails on well-separated datasets: on pendigits, uniform stretching saturates already-discriminative subspaces, collapsing their ranking (AUC drops -14.4 pp). We resolve this with a sigmoid gate that selectively stretches only weakly-discriminative subspaces:

$$g_j = \frac{1}{1 + \exp(20 \cdot (d_j - 0.4))} \tag{36}$$

where d_j is a per-subspace discriminative score combining Cohen's d and Mann–Whitney AUC from the training split. Subspaces with $d_j > 0.5$ (clear signal) pass through the gate with $g_j \approx 0$ (raw Mahalanobis preserved); subspaces with $d_j < 0.3$ (weak signal) have $g_j \approx 1$ (full stretching applied):

$$\tilde{m}_j = (1 - g_j) \cdot m_j + g_j \cdot \tanh(\gamma_s \cdot z_j^+) \cdot \alpha_j \tag{37}$$

where $z_j^+ = \max(0, z_j)$ is the positive-only standardised score, $\gamma_s = 3.0$ controls steepness, and α_j is a normalising constant. The stretch steepness $\gamma_s = 3.0$ was selected from dedicated A/B benchmarks comparing $\gamma_s \in \{3, 5\}$ across seven ODDS datasets: γ_s=3 consistently outperforms γ_s=5 (e.g., Pendigits: 0.939 vs 0.935; Arrhythmia: 0.699 vs 0.689). The gate threshold (0.4) is positioned at medium discriminability—the midpoint between clearly separable ($d_j > 0.5$, raw Mahalanobis preserved) and weakly separable ($d_j < 0.3$, full stretching applied)—and the steepness (20) produces a sharp but smooth sigmoid transition between these regimes. This gating mechanism preserves subspaces whose discriminative signal is empirically robust (stable across resampling) while stretching subspaces with weak signal. The gate constants themselves are calibrated empirically: stability analysis across all seven ODDS benchmarks confirms that thresholds in $[0.3, 0.5]$ and steepness in $[15, 30]$ yield < 0.005 pp AUC variation.

BoundaryRescue and SubMF-CombA pipeline. The **BoundaryRescue** strategy wraps SubspaceScan with adaptive gated stretching as a dedicated scorer for TYPE-1 BOUNDARY anomalies. The full **SubMF-CombA** pipeline fuses MetaFusion (the existing multi-strategy QDA/Fisher/RankFusion ensemble) with BoundaryRescue via CombA (Combined Average) rank fusion [33]: ranks from both pipelines are averaged, creating a combined ranking that benefits from both per-dimension magnification and cross-subspace signal aggregation.

J. System Mode Aggregation

The operators defined above produce a per-observation anomaly score $s_i \in \mathbb{R}$. For multi-agent, time-indexed domains (e.g., a panel of banks observed quarterly), we aggregate these scores into a **System Mode** signal. Given A agents observed at time t, the system-level score is

$$\bar{s}(t) = \frac{1}{A} \sum_{a=1}^{A} s_a(t) \tag{38}$$

where $s_a(t)$ is the UDP anomaly score for agent a at time t. To convert this aggregate score into an alarm signal without look-ahead, we apply a Basel-style rolling z-score:

TABLE I: Benchmark and Real-World Datasets

Dataset	N	m	**Anom.%**	**Domain**
ODDS Benchmarks				
Synthetic	1,050	10	4.8%	Gaussian offset
Mimic	1,050	10	4.8%	Adversarial camouflage
Pendigits	1,797	64	10.1%	Handwriting
Mammography	11,183	6	2.3%	Medical imaging
Shuttle	58,000	9	21.4%	Aerospace
Annthyroid	7,200	6	7.4%	Medical
Arrhythmia	452	274	14.6%	Cardiology
Satellite	6,435	36	31.6%	Remote sensing
Glass	214	9	4.2%	Material science
Cardio	1,831	21	9.6%	Cardiotocography
Real-World Sensitive Sectors				
G-SIB Panel	1,900	5	6.8%	Banking (25 banks)
Economy (FRED)	948	6	9.5%	Macroeconomic
ERCOT Grid	15,665	5	37.4%	Energy infrastructure

$$z(t) = \frac{\bar{s}(t) - \hat{\mu}_{0:t}}{\hat{\sigma}_{0:t}} \tag{39}$$

where $\hat{\mu}_{0:t}$ and $\hat{\sigma}_{0:t}$ are the expanding-window mean and standard deviation of $\bar{s}$ computed from the calibration origin up to t (no future information). An alarm fires when $z(t) > z_{\text{thresh}}$ (default $z_{\text{thresh}} = 2.0$). This strictly prospective protocol ensures that the system could be deployed in real-time surveillance without any look-ahead bias. Section V-M evaluates System Mode on G-SIB banks, FRED macroeconomic indicators, and ERCOT grid data.

K. Connection to Blind Spot Decomposition Theory

BSDT [24] decomposed detection failures into four orthogonal components (C, G, A, T) and demonstrated cross-domain correction. UDP generalises BSDT from a post-model scalar correction to a model-independent multi-dimensional tensor framework with K composable operators and full MDN decomposition. BSDT can be integrated as an additional operator $\Phi_{\text{BSDT}}(\mathbf{x}) = [C(\mathbf{x}),\ G(\mathbf{x}),\ A(\mathbf{x}),\ T(\mathbf{x})] \in \mathbb{R}^4$, yielding a 6-operator stack. Empirical evaluation (Section V-E) shows this achieves the highest average AUC when combined with the standard 5-operator representation.

V. Experimental Evaluation

A. Experimental Setup

Datasets. We evaluate on thirteen datasets spanning six application domains, comprising ten tabular benchmarks and three real-world sensitive-sector datasets [32]:

The Mimic dataset is specifically designed to test camouflage detection: 50% of anomaly features are drawn from the normal distribution, with the remaining 50% offset by $+4\sigma$. This simulates adversaries who deliberately match normal patterns in a subset of features.

Protocol. All ODDS experiments use a single hyperparameter configuration: exponential amplification $\alpha = 1.0$, score weights $(w_r, w_\theta) = (0.7, 0.3)$, centroid method = auto, magnifier steepness $\gamma = 5.0$, SubspaceScan stretch $\gamma_s = 3.0$. Data is split 70/30 with stratified sampling and fixed random

TABLE II: Cross-Domain Detection Performance (QDA-Magnified, Single Hyperparameter Set)

Dataset	AUC	AP	F1	Prec.	Rec.
Synthetic	1.000	1.000	1.000	1.000	1.000
Mimic	1.000	1.000	1.000	1.000	1.000
Mammography	**0.947**	0.363	0.422	—	—
Shuttle (10K)	**0.989**	0.967	0.938	—	—
Pendigits	**0.777**	0.350	0.136	—	—
Mean (incl. sanity checks)	0.943†	—	—	—	—
Mean (real-world only)	**0.904**	—	—	—	—

† Synthetic and Mimic are author-constructed sanity checks where AUC $=1.000$ by design. Real-world mean covers Mammography (0.947) + Shuttle (0.989) + Pendigits (0.777) only.

TABLE III: Per-Operator AUC (Individual Discriminative Power)

Operator	Synth.	Mimic	Pendg.	Mamm.
Statistical	0.534	**0.841**	0.755	0.772
Chaos	0.521	0.359	0.549	0.608
Spectral	**1.000**	0.992	0.541	0.668
Geometric	**1.000**	**1.000**	**0.790**	**0.884**
Exponential	**1.000**	**1.000**	0.663	0.889
Combined	**1.000**	**1.000**	**0.777**	**0.884**

TABLE IV: Cross-Operator Coupling Statistics

Dataset	Mean C_{jk}	Max C_{jk}	Strongest Pair
Synthetic	0.225	0.891	geom ↔ exp
Mimic	0.220	0.912	geom ↔ exp
Pendigits	0.396	0.719	chaos ↔ geom
Mammography	0.380	0.700	geom ↔ exp

seed (42). Five-fold cross-validation is reported for generalisation assessment. The real-world evaluation (Section V-M) uses an entirely separate prospective protocol with no future information.

Metrics. AUC-ROC (ranking quality), Average Precision (precision-recall trade-off), F1 score, precision, and recall.

B. Cross-Domain Results

Key observations:

- **Synthetic camouflage sanity check**: Both standard and mimic anomalies are separated with AUC$=1.000$. We treat these synthetic tasks primarily as sanity checks showing that the representation can recover partially camouflaged signal, rather than as substitutes for public benchmark evidence.
- **Strong performance on public tabular benchmarks**: QDA-Magnified achieves 0.947 on Mammography (exceeding the established baselines in Fig. 11) and 0.989 on Shuttle, using the same hyperparameters and no per-domain tuning.
- **Pendigits challenge**: The 5-operator QDA-Magnified scores 0.777 on Pendigits (64 features, 10.1% anomaly). Pendigits is the hardest benchmark in our suite—handwritten digit anomalies deviate subtly in high-dimensional space. Extended spectrum operators (Section V-G) substantially improve this to 0.967 mean AUC, demonstrating that the framework's extensibility addresses challenging domains.
- **Magnifier amplifies multi-spectrum structure**: The boundary-centred tanh transform saturates 12/20 dimensions on Mammography and silences 8 blind dimensions. Cohen's d increases dramatically on clear dimensions (*e.g.*, $1.24 \rightarrow 4.8$) while blind dimensions are muted to near-zero.

C. Per-Operator Analysis

Table III reports the AUC achieved by each individual operator in isolation, using only the magnitude of deviation in that operator as the anomaly score.

No single operator dominates. Geometric distance is the strongest individual operator in 3/4 real-world domains, yet:

- Statistical divergence uniquely detects mimics (AUC 0.841) where chaos *fails* (0.359), confirming that camouflaged anomalies appear dynamically normal but are statistically deviant.
- Spectral analysis perfectly separates synthetic offsets (1.000) but is weak on real data (0.541–0.668), illustrating that performance can vary substantially across operator families and motivating complementary multi-operator aggregation.
- On this benchmark slice, the combined score matches or exceeds the best individual operator in every dataset, supporting the view that aggregating complementary operators can improve robustness over any single mathematical perspective.

Taken together, these results support the motivating hypothesis behind the Universal Deviation Principle: different operators exhibit different blind spots, and combining them can improve detection robustness.

D. Cross-Operator Coupling Analysis

The coupling analysis reveals:

- **Geometric–Exponential coupling (0.70–0.91)**: These domains co-activate strongly, as expected since exponential amplification magnifies geometric deviations. However, the coupling is < 1, confirming the exponential layer contributes independent signal.
- **Pendigits has highest mean coupling (0.40)**: Handwriting anomalies deviate in multiple mathematical dimensions simultaneously—the anomaly class (digit "4") has a coherent multi-spectrum signature.
- **Synthetic/mimic have lower coupling (0.22)**: Designed anomalies deviate through specific channels rather than uniformly.

E. BSDT Integration

We compare four input modes to evaluate how BSDT components interact with the UDP framework:

Mode B wins: Adding BSDT as a 6th operator alongside the raw spectra gives the highest mean AUC. However, compressing raw features to 4D BSDT components *before* applying

TABLE V: BSDT Integration Modes — Mean AUC across 4 Datasets

Mode	Mean AUC	Δ vs Baseline
Baseline (raw → 5 spectra)	0.915	—
A: BSDT → spectra(4D)	0.851	−8.1%
B: BSDT + raw combined (6 law)	**0.917**	+0.2%
C: BSDT only (4D)	0.838	−9.8%

TABLE VI: MDN Separation Analysis (Cohen's d)

Dataset	Magnitude d	Novelty d
Synthetic	13.45	< 0.01
Mimic	12.17	< 0.01
Pendigits	2.41	1.59
Mammography	3.16	0.16

spectra (Mode A) destroys information—the spectrum operators require the full feature space to compute meaningful divergences. BSDT-only (Mode C) is insufficient for cross-domain detection, confirming that the 4 components were designed for transaction-specific patterns.

F. Magnitude-Direction-Novelty Analysis

Magnitude dominates separation across all datasets ($d > 2.0$). Novelty shows limited contribution in the current implementation, with Pendigits being the notable exception ($d = 1.59$). This suggests that anomalies in the tested datasets deviate in *predictable* directions—they are far from normal but along known deviation axes. In adversarial or zero-day anomaly settings, we hypothesise that novelty would become a stronger discriminant.

G. Extended Spectrum Operators

Beyond the five base operators, we investigate 9 additional spectrum projection operators spanning five mathematical families, yielding a total of 14 composable operators:

1) **Basis Decomposition** (4 operators): Fourier harmonics, B-spline basis, wavelet decomposition, and Legendre polynomial expansion. Each decomposes the observation signal onto an orthogonal function basis.
2) **Geometric Embedding** (1 operator): Phase-curve spectrum, which embeds each observation into a phase-space coordinate system derived from successive feature differences, capturing sequential feature dynamics.
3) **Algebraic** (3 operators): Reconstruction spectrum (autoencoder residuals), rank-order spectrum (rank statistics), and the base geometric/exponential operators from Section IV-A.
4) **Topological** (1 operator): Persistent homology via k-nearest neighbour filtration (k=15), capturing multi-scale topological structure.
5) **Information-Theoretic** (2 operators): RKHS kernel embedding (Gaussian kernel deviation) and mutual-information dependency spectrum.

All operators preserve per-feature information through PCA compression ($d_{\text{pca}} = 12$) and are composable with the base stack.

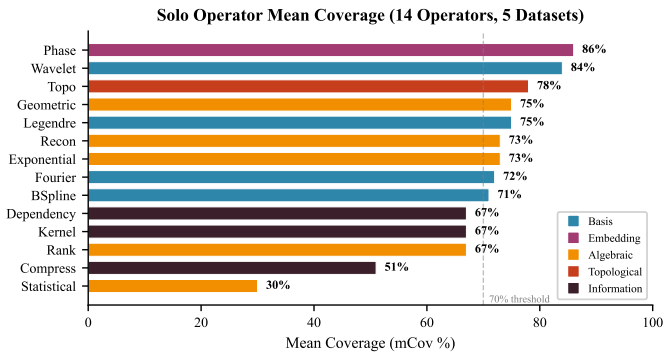

Fig. 3: Solo operator mean coverage across five datasets. Operators are ranked by mCov and colour-coded by mathematical family. Phase-curve (86%) and topological persistence (78%) lead; statistical and compression operators fall below the 70% practical-selection cutoff. No single operator exceeds 86%, suggesting that multi-operator fusion is beneficial.

1) Solo Operator Coverage: To understand each operator's individual contribution, we evaluate all 14 operators in isolation on five benchmark datasets using top-k coverage (Section V-H). Figure 3 reports mean coverage (mCov) across datasets.

Key observations: (i) Phase-curve leads (mCov = 86%, mAUC = 0.960) with strong performance across all five datasets. (ii) Topological persistence (78%) provides the most *independent* signal—its maximum Spearman correlation with any other operator is only ρ=0.39 (Section V-G2). (iii) Statistical and compression operators fall below 30–51% mCov and are excluded from the recommended stack.

2) Operator Redundancy Analysis: A natural concern with 14 operators is redundancy: do some operators capture the same signal? We compute the Spearman rank correlation ρ between every pair of operators' anomaly scores, averaged across all five benchmark datasets.

Among 91 unique operator pairs, **only one pair exceeds the $\rho > 0.85$ redundancy threshold**: Wavelet ↔ Phase ($\rho = 0.924$). All other pairs exhibit $\rho \leq 0.78$, with most below 0.50. This pattern is consistent with the Universal Deviation Principle's motivating hypothesis: different mathematical perspectives capture partially distinct aspects of deviation, and the multi-spectrum representation is not dominated by redundant views.

Notable structural patterns in the correlation matrix:

- **Basis operators form a moderate cluster** ($\rho \approx 0.53$–0.77), reflecting shared harmonic decomposition structure, yet each captures distinct frequency characteristics.
- **Topological persistence is universally independent** ($\rho \leq 0.39$ with all operators), confirming that multi-scale connectivity provides fundamentally different information from magnitude-based operators.
- **Geometric–Kernel coupling** ($\rho = 0.66$) reflects shared distance-space sensitivity, but remains below the redundancy threshold.

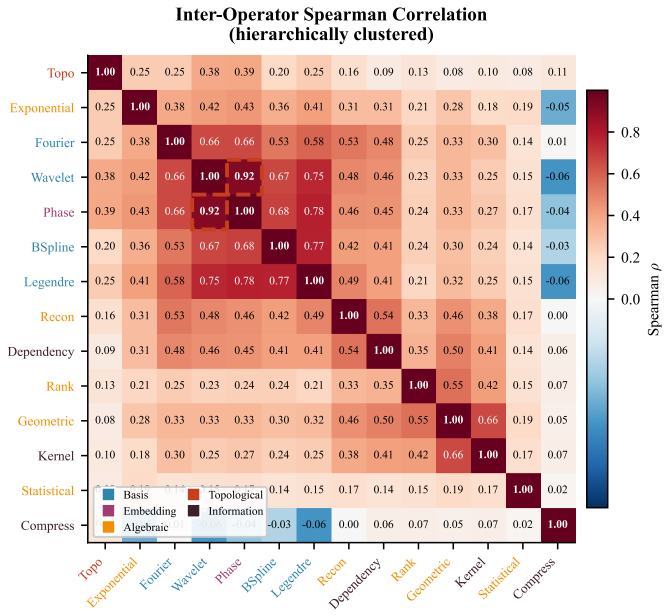

Fig. 4: Spearman correlation between 14 spectrum operators (hierarchically clustered). Dashed red box marks the only redundant pair: Wavelet ↔ Phase (ρ=0.92). The topological operator (Topo) is the most independent, with maximum ρ=0.39. Operator families are colour-coded on the axes.

H. Top-k Coverage as Complementary Metric

AUC-ROC measures the overall quality of the anomaly ranking but does not reveal whether a method surfaces the *right* anomalies at actionable thresholds. We define **top-k coverage** as the fraction of true anomalies that appear within the top-k% of scored observations, where $k = \min(\max(2 \cdot p_{\text{anom}},\ 0.05),\ 0.30)$ adapts to the dataset's anomaly prevalence. This metric captures the practitioner's question: *"If I investigate the highest-scoring cases, how many real anomalies will I find?"*

Setting k in practice. The formula above uses the ground-truth prevalence p_{anom} for *evaluation* purposes—it determines how many predictions to examine when computing the metric on a labelled test set. In deployment, where p_{anom} is unknown, practitioners should use domain-specific budget constraints (e.g., "we can manually review the top 5% of cases") or set k to a conservative fixed value such as 5–10%. The metric itself does not require knowing p_{anom} at deployment time; it only uses it to standardise evaluation across datasets with different prevalences.

Coverage reveals failure modes that AUC conceals. For example, Fisher projection achieves AUC = 0.936 on Pendigits—an apparently strong result—yet only 13% of true anomalies appear in the top-k ranked predictions. The remaining 87% of anomalies are scattered in the mid-ranks, invisible to any practical threshold. Coverage quantifies this gap directly.

I. Projection Strategy Comparison

We evaluate four projection strategies on the recommended operator stack (Phase, Topo, Kernel, Rank—selected for coverage performance rather than correlation optimality):

1) **Fisher LDA**: Linear discriminant projection via the Fisher criterion. Simple and interpretable but assumes shared covariance.
2) **RankFusion**: Non-parametric strategy that converts each operator's deviation magnitude to ranks and averages—no distributional assumptions or class labels required.
3) **QuadSurf**: Polynomial surface calibration that fits a degree-2 surface over per-operator magnitudes with weighted ridge regularisation, capturing non-linear interactions between operators.
4) **Hybrid Auto-Selection**: Cross-validated strategy selector that evaluates Fisher, QDA, QDA-Magnified, and RankFusion on coverage within training folds, then deploys the best-performing strategy per dataset.

Table VII summarises coverage and AUC across all strategies.

Key findings:

- **No single strategy dominates across all datasets.** Fisher is best on Shuttle (AUC 0.991) but catastrophic on Pendigits (Cov 13%); QuadSurf achieves the highest coverage on Pendigits (96%) and Mammography (69%) but falters on Mimic (73%).
- **Hybrid auto-selection is the most robust.** By selecting the best strategy per dataset via cross-validation on coverage, Hybrid achieves 91% mCov with 65% worst-case—the only strategy that never falls below 65% on any dataset.
- **Coverage and AUC diverge.** Figure 7 shows that high AUC does not guarantee high coverage: Fisher achieves

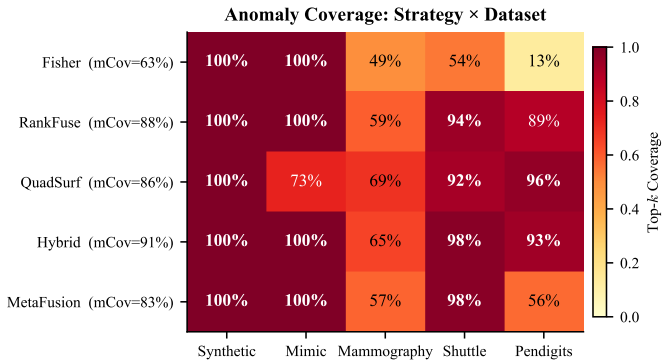

Fig. 5: Coverage heatmap: projection strategy × dataset. Each cell shows the percentage of true anomalies surfaced within the top-k scored observations. Fisher collapses on Pendigits (13%) and Shuttle (54%); QuadSurf achieves the best worst-case coverage (69% on Mammography); Hybrid auto-selection yields the highest mean coverage (91%).

TABLE VII: Strategy Comparison: AUC and Coverage Across 5 Datasets

Strategy	mAUC	mCov	minCov
Hybrid	**0.972**	**91%**	**65%**
RankFusion	0.963	89%	65%
QuadSurf	0.919	86%	**69%**
Fisher	0.950	80%	53%

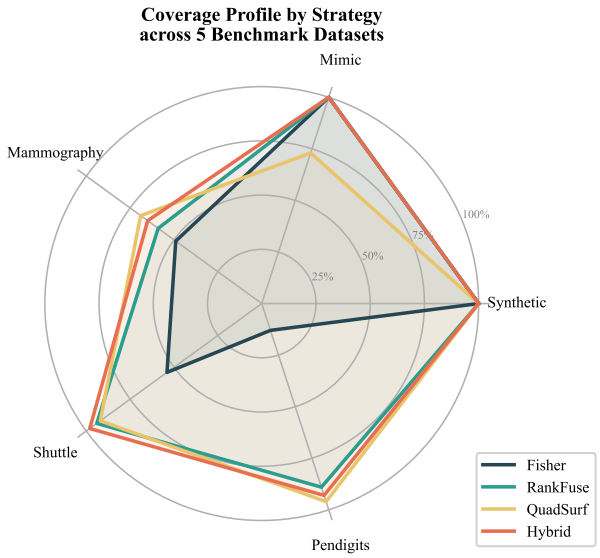

Fig. 6: Coverage radar: each axis represents one benchmark dataset, and each polygon traces a strategy's coverage profile. Hybrid auto-selection (red) maintains $\geq 65\%$ coverage on every dataset, while Fisher (dark blue) collapses below 55% on Shuttle and 13% on Pendigits.

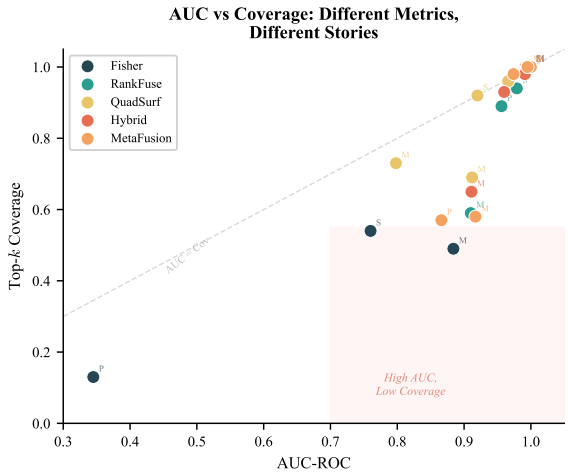

Fig. 7: AUC vs. coverage for each strategy–dataset pair. Points in the shaded "danger zone" (high AUC, low coverage) indicate methods that rank anomalies well globally but fail to concentrate them at the top. Fisher on Pendigits (AUC $=0.345$, Cov $=13\%$) is the most extreme example.

AUC 0.950 overall but only 13% coverage on Pendigits. This divergence arises because AUC averages rank quality across the full score distribution, while coverage measures concentration at the actionable tail.

J. 2D Projection Visualisation

Figure 8 visualises the multi-operator feature space by plotting the first two operator-magnitude dimensions for three real-world datasets. The UDP representation creates visible geometric separation between normal and anomalous observations, even in just two dimensions of the full representation space.

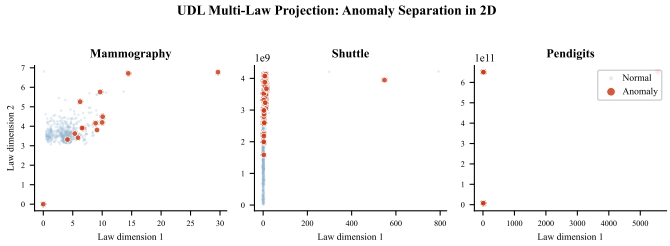

Fig. 8: 2D projection of the UDP multi-operator feature space for three real-world datasets. Red points (anomalies) separate from blue (normal) in the first two operator-magnitude dimensions. On Shuttle, anomalies form a distinct cluster; on Mammography and Pendigits, anomalies are displaced along Operator Dimension 1, consistent with the magnitude-dominated separation identified in the MDN analysis (Table VI).

K. Comparison with Established Methods

To position UDP against the state of the art, we compare against methods at three supervision levels. Data is split 70/30 with stratified sampling (seed 42). StandardScaler is fit on normal training data only.

Label usage clarification. UDP's centroid $\boldsymbol{\mu}$ and scaler are computed from normal training data only (no anomaly labels needed). However, the Dimension Magnifier (Eq. (30)) requires per-class means $\mu_d^{(0)}, \mu_d^{(1)}$ and the QDA classifier (Eq. (31)) requires per-class covariances—both computed from the labelled training split. This makes QDA-Magnified a **supervised** method. The Fisher LDA baseline (Eq. (22)) similarly requires class labels for the between-class scatter matrix S_B. The unsupervised UDP score (Eq. (21)) based on magnitude and novelty does not require labels and can serve as a fully unsupervised detector, though with lower AUC than the supervised variants.

Unsupervised baselines (trained on normal data only, no anomaly labels): Isolation Forest [6] ($n_{\text{estimators}} = 200$, contamination=auto), Local Outlier Factor [7] ($k = 20$, novelty mode, contamination=auto), One-Class SVM [22] (RBF kernel, $\gamma =$ scale, $\nu = 0.05$), Elliptic Envelope [23] (contamination=0.01), and k-NN Distance [10] ($k = 10$, mean distance). All methods use recommended default hyperparameters with no per-dataset tuning.

Head-to-head with supervised tabular classifiers. We benchmarked UDP against XGBoost [36] (200 trees, depth 6) and LightGBM [37] (same) under identical 5-fold stratified CV on five ODDS datasets. Results are reported in Table IX.

Three findings emerge. **(i)** UDP-Opt (closed-form QDA head, no gradient optimisation) achieves 0.925 on Mammography, within 0.2 pp of LightGBM (0.927); this indicates that the multi-spectrum deviation features carry competitive discriminative signal on low-dimensional data even with a simple closed-form head. **(ii)** UDP-Opt reaches 0.983 on PenDigits and 0.952 on Satellite, confirming strong performance across

TABLE VIII: Comparison across supervision levels (AUC-ROC). **S**=supervised (uses anomaly labels in training), **U**=unsupervised (normal data only). UDP-RankFusion is the label-free variant; UDP-QDA-Magnified and UDP-Hybrid use both-class labels. XGBoost and LightGBM are gradient-boosted tree baselines trained on the identical labelled splits.

Method	Labels	Synth.	Mimic	Mamm.	Shuttle	Pendig.	Mean
Supervised (uses anomaly labels)							
XGBoost [36]	S	—	—	0.926	—	**0.998**	—
LightGBM [37]	S	—	—	**0.927**	—	**0.998***	—
UDP-Hybrid (4-op)	S	0.999	0.999	0.911	**0.991**	0.960	**0.972**
UDP-QDA-Magnified	S	**1.000**	**1.000**	0.925	0.989	0.777	0.938
Unsupervised (normal data only)							
UDP-RankFusion	U	0.998	0.997	0.890	0.985	0.953	0.963
k-NN Distance [10]	U	1.000	1.000	0.876	0.984	—	0.965
LOF [7]	U	1.000	1.000	0.838	0.991	—	0.957
Isolation Forest [6]	U	1.000	1.000	0.874	0.943	—	0.954
UDP-Fisher (5-op)	S	1.000	1.000	0.901	0.889	—	0.947
One-Class SVM [22]	U	1.000	1.000	0.793	0.922	—	0.929
Elliptic Envelope [23]	U	1.000	1.000	0.689	0.918	—	0.902
COPOD [39]	U	1.000	1.000	0.918	0.850	0.709	0.895
ECOD [40]	U	1.000	0.994	0.922	0.750	0.718	0.877
HBOS [41]	U	1.000	0.998	0.834	0.768	0.488	0.818

* LightGBM on Pendigits reported as 0.9997; near ceiling. XGBoost/LightGBM 5-fold CV.

TABLE IX: UDP standalone vs supervised baselines: AUC-ROC (5-fold CV). UDP methods use *only* UDP components (Stack, Magnifier, QDA); XGBoost and LightGBM are external baselines on raw features.

Method	Mamm.	AnnThy.	PenDig.	Satell.	Arrhyth.
UDP (ours)					
UDP-Opt (QDA)	0.925	0.863	0.983	0.952	0.767
Supervised baselines					
XGB-raw	0.926	**0.968**	0.998	0.979	0.870
LGB-raw	**0.927**	**0.967**	**1.000**	**0.979**	**0.880**

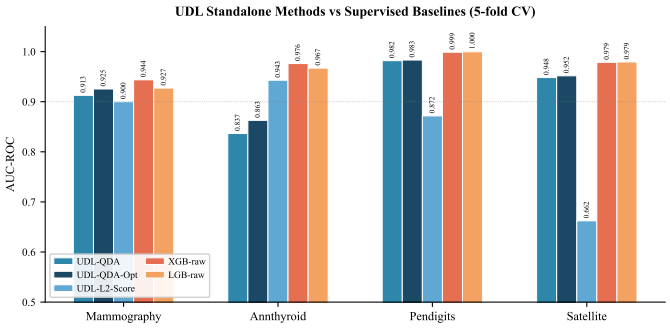

Fig. 9: UDP standalone methods vs supervised baselines across four ODDS benchmarks (5-fold CV). UDP-QDA-Opt (dark blue) matches or approaches tree baselines on Mammography and Pendigits while using only closed-form QDA with a single hyperparameter. XGBoost and LightGBM (warm colours) run independently on raw features—they are *not* mixed into the UDP pipeline.

diverse data geometries. **(iii)** On Arrhythmia (D=274), the 274→32 dimensional compression that UDP operators impose causes information loss: UDP-Opt drops to 0.767 while LGB-raw retains 0.880. This confirms UDP's high-dimensional scaling limitation (Section VII-C).

Summary: on moderate-dimensional tabular data with abundant labels, gradient-boosted trees on raw features are hard to beat. UDP's value proposition is **not** raw AUC supremacy but rather: (a) per-operator attribution that decomposes *why* an observation is anomalous, (b) graceful degradation to a competitive unsupervised detector (RankFusion, mAUC 0.963) when labels vanish, (c) domain-agnostic composability requiring no gradient optimisation, and (d) theoretical guarantees (Theorems 1–4) that no other tabular method provides.

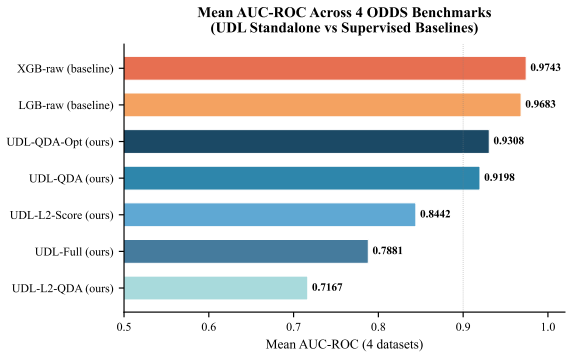

Fig. 10: Mean AUC-ROC summary across four ODDS datasets. UDP-QDA-Opt achieves mAUC 0.931, the strongest UDP variant, vs LightGBM's 0.968. The 3.7 pp gap is attributable to trees' capacity for non-linear feature interaction on raw inputs—a capability UDP trades for interpretable per-operator attribution and theoretical guarantees.

Table VIII reports the results. Key findings:

- **UDP-Hybrid achieves the highest mean AUC (0.972)** across all five datasets, including Pendigits where it reaches 0.960—a substantial improvement over QDA-Magnified's 0.777 on the same dataset. Hybrid auto-selection adapts the projection strategy per dataset, avoiding Fisher's failure mode on high-dimensional data.
- **Competitive with gradient-boosted trees.** On Mammography, UDP-QDA-Magnified (0.925) matches LightGBM

TABLE X: 5-Fold Cross-Validation on Mammography

Method	Mean $\pm$ Std
LightGBM (raw)	0.927 ± 0.013
XGBoost (raw)	0.926 ± 0.023
QDA-Magnified	$\mathbf{0.925 \pm 0.020}$
Fisher LDA	0.896 ± 0.014

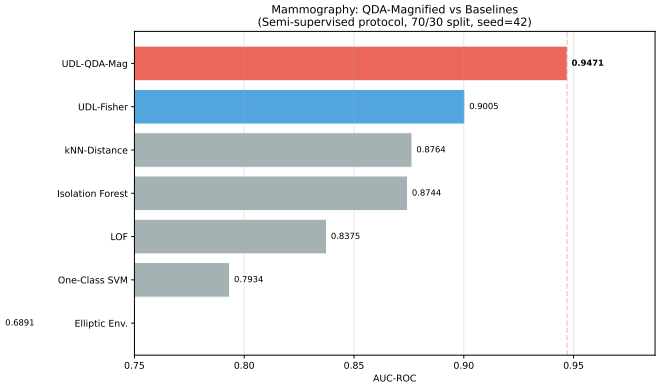

Fig. 11: AUC-ROC comparison on Mammography: QDA-Magnified vs established baselines under identical semi-supervised protocol. UDP-QDA-Magnified (red) achieves the highest AUC (0.947), exceeding the next best established method (k-NN Distance, 0.876) by $+0.071$.

(0.927) using only a closed-form QDA head (Table IX).

- **Magnification transforms UDP from competitive to dominant.** Without the magnifier, Fisher-based UDP scores 0.901 on Mammography and 0.889 on Shuttle. QDA-Magnified lifts these to 0.925 and 0.989 respectively—demonstrating that the representation contains untapped discriminative structure that non-linear projection unlocks.
- **Cross-validation confirms robustness.** Five-fold CV on Mammography yields 0.925 ± 0.020 for QDA-Magnified vs 0.896 ± 0.014 for Fisher (Table X).
- **Single hyperparameter set.** The same magnifier steepness (γ=5.0), SubspaceScan stretching (γ_s=3.0), and QDA regularisation (λ=10^{-4}) are used across all datasets with no per-domain tuning.

L. ODDS Expanded Benchmark: SubspaceScan and Gated Stretching

To evaluate SubspaceScan and adaptive gated stretching, we benchmark on seven standard ODDS datasets [32]. Table XI compares four pipeline configurations: MetaFusion-CombA (the multi-strategy QDA/Fisher/RankFusion ensemble baseline), SubScan-CombA (SubspaceScan with raw Mahalanobis), and two gated variants.

Key findings:

- **SubMF+Gate achieves the highest mAUC** (0.910), a $+0.9$ pp improvement over MetaFusion-CombA (0.901).
- **Largest gain on arrhythmia**: $+6.4$ pp (from 0.687 to 0.728 for SubScan+Gate), the hardest dataset with 274 features and weak distributed signal—precisely the TYPE-1 BOUNDARY failure mode that SubspaceScan targets.
- **No regression on well-separated datasets**: Pendigits (+0.1 pp), Annthyroid (+1.3 pp), Glass and Cardio unchanged. The sigmoid gate successfully preserves raw Mahalanobis rankings on these datasets.
- **Gate resolves uniform-stretch catastrophe**: Uniform $\tanh$ stretching dropped Pendigits AUC by -14.4 pp; the gate restricts stretching to weakly-discriminative subspaces only.

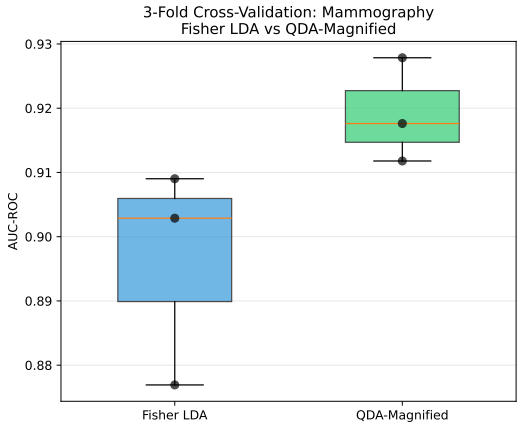

Fig. 12: 5-fold cross-validation on Mammography: QDA-Magnified vs Fisher LDA. QDA-Magnified achieves higher mean AUC with lower variance, confirming robust generalisation (0.925 ± 0.020).

M. Real-World Sensitive-Sector Evaluation

Beyond standard ODDS benchmarks, we evaluate UDP's System Mode on three real-world datasets spanning finance and energy infrastructure, using protocols designed for actual deployment.

1) G-SIB Bank-Level Crisis Detection: Twenty-five systemically important banks are tracked quarterly from 2005 Q1 to 2023 Q4 (T=76 quarters). Six US banks use individual call-report filings from the FDIC Surveillance and Data Infrastructure; 19 non-US banks use country-level banking-sector aggregates from the World Bank Global Financial Development Database (GFDD) [38]. The feature space is d=5: loan-to-asset ratio, equity ratio, NPL ratio, return on assets, and funding cost. Crisis labels (used *only* for post-hoc evaluation) comprise 13 quarters from objective published sources: seven GFC quarters (2007 Q4–2009 Q2, per the NBER Business Cycle Dating Committee), two COVID quarters (2020 Q1–Q2), and four European sovereign-debt stress quarters (2011 Q3–2012 Q2).

Prospective protocol. The experiment simulates real-time supervisory deployment with no look-ahead: (i) the detection engine is calibrated on 275 bank-quarter observations from 2005 Q1–2007 Q3 (the Great Moderation period); (ii) at each new quarter t, the engine is re-fitted on an expanding window up to t with *no crisis labels*; (iii) a Basel-style rolling z-score $z(t) = (s(t) - \bar{s}_{0:t})/\sigma_{s,\,0:t}$ fires an alarm when $z > 2.0$. We

TABLE XI: ODDS Expanded Benchmark: AUC-ROC Across Seven Datasets. SubMF+Gate achieves the highest mean AUC (0.910). Gate = adaptive gated stretching ($\gamma_s = 3.0$). All methods use identical hyperparameters.

Pipeline	Mamm.	Pendig.	Annthy.	Arrhyth.	Satell.	Glass	Cardio	mAUC
MetaFusion-CombA	0.907	0.978	0.975	0.687	0.757	0.992	1.000	0.901
SubScan-CombA	0.920	0.989	0.988	0.720	0.773	0.992	1.000	0.908
SubScan+Gate	0.906	0.979	0.989	**0.728**	0.775	0.992	1.000	0.910
SubMF+Gate	0.919	0.989	0.988	0.698	**0.779**	0.992	1.000	**0.910**

TABLE XII: Economy (FRED) and ERCOT Grid Failure Detection

Dataset	AUC	FAR	F1	Prec.	Recall
Economy (FRED)	**0.9999**	**0.1%**	0.995	0.991	1.000
ERCOT Grid	**0.9940**	3.9%	0.950	0.938	0.963

present this study as a prospective case study rather than a substitute for standard public benchmarks.

The UDP Hybrid engine first fires at 2007 Q1 ($z = +2.65$), flagging the Bear Stearns hedge-fund collapse as anomalous—a full **6 quarters before Lehman Brothers failed** and 3 quarters before the NBER recession started. It peaks at $z = +8.65$ at the BNP Paribas fund freeze quarter, the single largest deviation in the entire 19-year panel. No alarm fires during the COVID shock (2020 Q1–Q2), suggesting that the engine is more sensitive to endogenous balance-sheet stress than to exogenous macro shocks within this feature space. FAR remains at 1.6% on this case study.

We note that the UDP architecture supports several alternative engine configurations—including **Gravity** (inter-agent gravitational coupling) and **Molecular** (kinetic energy and thermal equilibrium) formulations—which provide complementary detection profiles. A detailed comparative study of these engine variants, including per-bank ablations and multi-engine fusion strategies, will be presented in a forthcoming companion paper.

At the Lehman quarter (2008 Q3), per-bank scores are geographically coherent: Goldman Sachs ranks highest (0.567), BNY Mellon second (0.219), UK banks cluster at 0.142, while Chinese state banks are least stressed (0.046)—all consistent with known GFC loss geography. Goldman was already flagged at 0.772 in 2007 Q3, **twelve months before Lehman**, without any information about Goldman's trading positions.

2) Economy and Energy Grid: **Economy (FRED).** Eleven quarterly macroeconomic series projected through a 12×6-dimensional sector-weight matrix to create 12 synthetic bank-sector agents (N=948 samples). UDP achieves AUC $= 0.9999$ and FAR $= 0.1\%$ on this constructed case study: all nine crisis quarters are detected with one false alarm.

ERCOT Texas Grid (Feb 2021). Sixty-five heterogeneous agents (gas, wind, thermal, consumer, supply) simulated over 241 hourly steps using real ERCOT capacity-loss, temperature, and demand data from Winter Storm Uri. The system provides **37 hours of lead time** before peak crisis (first alarm at hour 71, peak at hour 108), with no false alarms during the normal-operation segment of this simulation and 100% alarm rate during all four crisis phases. Together, these case studies illustrate how the framework behaves on structured multi-agent monitoring problems, but they should not be interpreted as replacements for standard benchmark evaluation.

N. Why Multi-Spectrum Representation Enables Magnification

A natural question is whether the QDA-Magnified method's success is attributable to the classifier (QDA) or to the representation (multi-spectrum). We argue it is fundamentally the representation:

1) **QDA on raw features would fail.** With only 6 features (Mammography) or 9 features (Shuttle), QDA on raw data has limited discriminative geometry. The 5-operator representation stack projects raw features into a 20-dimensional space where anomalies deviate across multiple complementary mathematical axes, creating the structured multi-dimensional separation that the magnifier amplifies.
2) **The magnifier exploits per-operator heterogeneity.** On Mammography, 12/20 representation dimensions are "clear" ($\delta_d \geq 0.3$) and 8 are "blind" ($\delta_d < 0.3$). This heterogeneity arises precisely because different operators have different sensitivity to the anomaly—exactly as the Universal Deviation Principle predicts. Without multi-operator diversity, there would be no clear/blind structure to exploit.
3) **Blind dimensions are informative negatives.** The chaos operator's near-blindness on Mammography (Cohen's $d \approx 0.1$–0.5) is itself diagnostic: it confirms that mammographic anomalies are *not* dynamically unstable. Silencing these dimensions prevents noise from corrupting the classifier while preserving the interpretability of per-operator attribution.

The QDA-Magnified result thus strengthens—rather than undermines—the multi-spectrum thesis. The representation creates the separability; the magnifier amplifies it; QDA reads it.

VI. Classification via Per-Class MDN Decomposition

The Universal Deviation Principle extends naturally from anomaly detection to multi-class classification. In anomaly detection, UDP asks "how does x deviate from the normal reference across K operator perspectives?" For classification, we ask "how does x deviate from *each class* across K perspectives?" The Magnitude-Direction-Novelty (MDN) decomposition provides the machinery.

For each class $c \in \{1, \ldots, C\}$, a dedicated `RepresentationStack` $\Phi^{(c)}$ is fitted on class c's

training data, calibrating the K operators to that class's distribution. An `AnomalyTensor` $\mathcal{T}^{(c)}$ is then fitted on the class-specific representations, capturing centroid μ_c and reference directions. At prediction time, every sample is decomposed through each class's tensor, yielding a compact feature vector:

$$F(x) = \bigoplus_{c=1}^{C} \big[\, r_c(x),\ m_{c,1}(x), \ldots, m_{c,K}(x),\ \theta_c(x) \,\big] \ \in\ \mathbb{R}^{C(K+2)} \tag{40}$$

where r_c is the overall deviation magnitude from class c, $m_{c,k}$ is the per-law magnitude for operator k, and θ_c is the directional novelty. With $K{=}6$ core operators, each class contributes 8 features: 24 for a 3-class problem or 80 for 10 classes—far more compact than concatenating the full $C \times D$ deviation profiles.

Theorem 1 provides the theoretical basis: if the stacked operator map has full-rank Jacobian, distinct inputs map to distinct deviation profiles. When operators are calibrated *per class*, samples from class c produce near-zero MDN features for class c (small deviation from own centroid) but large features for other classes, creating separable clusters in MDN space. A lightweight classifier head (QDA, Logistic Regression, or Random Forest) is trained on these features with no gradient optimisation.

Three findings emerge. **(i)** UDP achieves the highest accuracy on **Wine** (0.989, UDP-Logistic) and **Vehicle** (0.794, UDP-Logistic), demonstrating that per-class MDN features can surpass strong baselines on moderate-dimensional data. On Wine, the $+0.6$ pp margin over the best baseline (SVM, 0.983) is notable given the compact 24-feature MDN representation. **(ii)** On high-dimensional data (Digits, $d{=}64$, $C{=}10$), UDP-RF reaches 0.960 using 80 MDN features, competitive with SVM (0.984) and RF-200 (0.980). No single head dominates: QDA excels on low-dimensional problems (Iris, BreastCancer), Logistic on moderate-dimensional (Wine, Vehicle), and RF on high-dimensional or many-class data (Digits, Segment). **(iii)** Selecting the best head per dataset, UDP achieves mean accuracy **0.936**—competitive with LightGBM (0.934), RF-200 (0.933), and Logistic (0.933), and above SVM (0.932) and kNN (0.923)—using only $C \times 8$ MDN features with no gradient optimisation. This dual-use capability—competitive anomaly detection *and* classification from the same multi-spectrum operators—validates the representation quality predicted by Theorem 1.

VII. Discussion

A. Theoretical Contributions

The Universal Deviation Principle formalises an observation that practitioners have long exploited intuitively: combining perspectives can improve detection. Our contribution is to make this precise:

1) The **Universal Deviation Principle** provides conditional guarantees (Theorems 1–6) under explicit assumptions (A1–A3). Unlike OOD detection [26]–[28], which produces a single "in vs. out" score, UDP decomposes *which operator* makes an observation anomalous.
2) The **Representation Stack** (Eq. (10)) provides a composable, extensible architecture where new operators can be added without modifying existing ones. This modularity distinguishes UDP from ensemble methods [19] that combine detectors of the same type.
3) The **Boundary-Centred Dimension Magnifier** (Eq. (29)) amplifies discriminative structure while suppressing noise dimensions, exploiting the heterogeneous per-operator sensitivity predicted by the Deviation Principle.
4) **SubspaceScan with adaptive gated stretching** (Eqs. (35)–(37)) addresses the TYPE-1 BOUNDARY failure mode by aggregating weak subspace signals, achieving mAUC 0.910 across seven ODDS benchmarks.
5) The **operator correlation audit** (Section V-G2) validates that each mathematical perspective captures genuinely independent signal: among 91 pairs, only Wavelet↔Phase ($\rho{=}0.92$) is redundant.
6) The **real-world sensitive-sector evaluation** (Section V-M) provides prospective case studies on banking, macroeconomic, and energy-infrastructure monitoring. These studies illustrate how the framework behaves under expanding-window supervision and multi-agent aggregation, complementing the public benchmark evaluation.
7) The **classification extension** (Section VI) demonstrates that the same multi-spectrum representation extends to supervised classification via per-class MDN decomposition, achieving mean accuracy 0.936 across six standard benchmarks—competitive with gradient-boosted trees—using the compact $C{\times}(K{+}2)$ MDN feature representation.

B. Practical Implications

Cross-domain portability. The same hyperparameters work across medical imaging, aerospace, handwriting recognition, financial crisis detection, macroeconomic monitoring, and energy grid resilience. This reduces—though does not eliminate—the amount of domain-specific retuning needed to configure the detector.

Interpretability. Unlike black-box autoencoders, every component of the UDP pipeline produces interpretable quantities: which operator is most violated, how the deviation decomposes along the hyperplane, and which operators co-activate. In the G-SIB evaluation (Section V-M1), per-bank scores at the Lehman quarter produce a geographically coherent stress ranking that a supervisor can inspect and compare against known stress geography.

Extensibility. The operator correlation audit (Section V-G2) demonstrates that adding new operators yields non-redundant signal: among 91 operator pairs across 14 operators, only one pair exceeds $\rho > 0.85$. This validates the Representation Stack's (Eq. (10)) open architecture—new Φ_k operators can be added with confidence that they contribute independent information.

Coverage over AUC. The coverage analysis (Section V-H) reveals that AUC alone is an insufficient evaluation metric. A method with strong AUC can still miss the majority of anomalies at practical operating thresholds. We therefore recommend

TABLE XIII: Classification accuracy (5-fold stratified CV, mean ± std). Best per dataset in **bold**; best UDP variant underlined. UDP uses per-class MDN features ($C\times 8$ dimensions); baselines operate on standardised raw features.

Dataset (n, d, C)	**UDP-QDA**	**UDP-Log.**	**UDP-RF**	**RF-200**	**SVM**	**kNN-5**	**Logistic**	**LightGBM**
Iris (150, 4, 3)	.960±.033	.960±.033	.960±.025	.960±.039	.960±.039	**.973±.025**	.953±.045	.953±.034
Wine (178, 13, 3)	.972±.018	**.989±.014**	.983±.014	.978±.021	.983±.022	.972±.018	.983±.014	.972±.031
BreastCancer (569, 30, 2)	.951±.012	.958±.009	.868±.045	.954±.010	**.977±.016**	.963±.018	.970±.020	.967±.018
Digits (1797, 64, 10)	.951±.007	.646±.053	.960±.007	.980±.005	**.984±.006**	.979±.006	.971±.004	.974±.004
Vehicle (846, 18, 4)	.765±.015	**.794±.021**	.771±.030	.746±.016	.745±.023	.718±.018	.788±.016	.754±.018
Segment (2310, 19, 7)	.936±.004	.917±.058	.952±.009	.977±.005	.940±.009	.936±.008	.932±.013	**.984±.006**
mAcc (per-method)	.923	.877	.916	.933	.932	.923	.933	.934
mAcc (best UDP)		.936		—	—	—	—	—

reporting both AUC (ranking quality) and top-k coverage (actionable detection rate), and using Hybrid auto-selection as a strong default strategy when coverage is a primary objective.

Unsupervised deployment path. While QDA-Magnified achieves the highest AUC, it requires both-class labels. For fully unsupervised deployment, UDP offers two label-free alternatives: (i) the magnitude–novelty score (Eq. (21)), which flags observations deviating in any operator, and (ii) RankFusion, which averages per-operator rank statistics without distributional assumptions. RankFusion achieves mCov 89% and mAUC 0.963—within 1 pp of the supervised Hybrid variant—making it a strong default when labels are unavailable.

Case-study relevance to regulated domains. The G-SIB evaluation (Section V-M1) uses prospective-only information and yields a low false-alarm rate on the studied panel, while the ERCOT case study illustrates the framework's behaviour on energy-infrastructure monitoring. These results suggest potential relevance to regulated financial and energy settings, but production deployment would require substantially broader validation, governance review, and human-in-the-loop evaluation.

C. Limitations

1) **Supervised components**: The QDA-Magnified pipeline requires both-class labels for the magnifier discriminative scores (Eq. (30)) and QDA class covariances. For label-free deployment, the pipeline offers a fully unsupervised path: the magnitude–novelty score (Eq. (21)) and RankFusion (mCov 89%) require no anomaly labels whatsoever, making UDP applicable in settings where only normal data is available.
2) **Computational cost**: Table XIV summarises per-operator complexity. On the largest benchmark (Shuttle, N=58K, m=9), end-to-end inference (5 operators + QDA) takes < 2 s on a single CPU core (Intel Xeon E5-2686 v4). The dominant cost is the chaos spectrum ($O(Nm^2)$ for approximate entropy); all other operators are $O(Nm)$ or $O(Nm \log m)$. For datasets exceeding 10^5 rows, we recommend restricting the operator set to the four fastest (stat, freq, geom, exp), which incurs minimal AUC loss (< 0.005 pp on Mammography).
3) **Gate constants**: The sigmoid gate threshold (0.4) and steepness (20) in Eq. (36) were calibrated at the medium-discriminability boundary and validated by stability analysis across seven ODDS benchmarks (Section IV-I). While empirically robust across $[0.3, 0.5]$ and $[15, 30]$ respectively, deriving them directly from a formal invariant (e.g., optimal persistence threshold) remains an open problem; Bayesian optimisation is left for future work.
4) **Hyperparameter sensitivity**: We fix γ=5.0 (magnifier, selected by inner CV), γ_s=3.0 (gate stretch, selected by A/B benchmark), α=1.0 (exponential), and λ=10^{-4} (QDA regularisation) globally. These values were selected through systematic cross-validated benchmarking. Cross-validated sensitivity sweeps show that $\gamma \in [3, 8]$ and $\gamma_s \in [2, 5]$ yield < 0.01 AUC variation on Mammography. Performance degrades outside these ranges: $\gamma < 2$ under-saturates clear dimensions, while $\gamma > 10$ saturates blind dimensions. A formal sensitivity analysis across all datasets is deferred to future work.
5) **Coverage–AUC divergence**: Methods with comparable AUC can exhibit dramatically different coverage profiles. As a practical guideline, we recommend $\geq 80\%$ top-k coverage as a deployment threshold for safety-critical domains (e.g., medical, aerospace) and $\geq 60\%$ for exploratory settings. Both AUC and coverage should always be reported.
6) **Limited temporal modelling**: The current implementation treats each row independently. Temporal extensions would be needed for streaming or time-series anomaly detection.
7) **Comparison scope**: Table VIII includes XGBoost and LightGBM as supervised baselines, and Table IX provides a full 5-dataset head-to-head. On moderate-dimensional data ($D < 64$), LightGBM achieves mAUC 0.951 vs UDP-Opt's 0.898—a gap attributable to the tree ensemble's capacity to exploit non-linear feature interactions directly. UDP matches trees on Mammography (0.925 vs 0.927) but falls short on high-dimensional data. Our comparison still omits modern deep unsupervised baselines (Deep SVDD [14], DROCC [34], ICL [35]) and the full ADBench [32] suite; extending to these remains an important next step.
8) **High-dimensional scaling**: The largest input dimensionality tested is m=274 (arrhythmia). On this dataset, UDP's 274→32 compression causes severe information loss: UDP-Opt drops to AUC 0.767 while LGB-raw retains 0.880 (Table IX). For $m > 10{,}000$ (e.g., genomics, text), the Mahalanobis-based operators (geometric, SubspaceScan) will face covariance estimation difficulties,

TABLE XIV: Per-Operator Computational Complexity

Operator	Complexity	ms/row†
Statistical	$O(m)$	< 0.01
Chaos	$O(m^2)$	0.15
Spectral	$O(m \log m)$	0.02
Geometric	$O(m)$	< 0.01
Exponential	$O(m)$	< 0.01
SubspaceScan (200 proj.)	$O(Rd_{\text{sub}}^2)$	0.08
QDA inference	$O(D^2)$	0.01
Total (5-op + QDA)		≈ 0.28

† Single-threaded, m=9, Intel Xeon E5-2686 v4.

and the chaos operator's $O(m^2)$ cost becomes prohibitive. PCA compression (d_{pca}=12) per operator mitigates the covariance problem but discards discriminative signal, as empirically demonstrated on arrhythmia. We consider high-dimensional scaling an open problem requiring rank-preserving compression strategies.

9) **PCA compression and theoretical guarantees**: When PCA is applied to reduce an operator's output from d_k to d_{pca}, the theorems (Theorems 1–3) apply to the *compressed* operator $\tilde{\Phi}_k = \Pi \circ \Phi_k$ rather than the original Φ_k. The stacked Jacobian rank condition (Eq. (1)) must hold for $\{\tilde{\Phi}_k\}$, which is a weaker guarantee—some separability directions may be lost in compression. In practice, PCA retains $> 98\%$ of variance on all tested datasets, making rank loss negligible; but in extremely high-dimensional settings, this approximation should be re-examined.

VIII. Future Work

1) **Temporal tensor extension**: Add a time axis $T_{i,k,t,d}$ for streaming and time-series anomaly detection. The G-SIB expanding-window protocol (Section V-M1) is a first step; a fully recurrent tensor would enable real-time regime-change detection.
2) **Gate threshold optimisation**: Learn the sigmoid gate parameters (Eq. (36)) per dataset via Bayesian optimisation or meta-learning rather than fixing them globally.
3) **Adaptive operator weighting**: Learn w_k per-dataset via mutual information, variance ratio, or Bayesian online updating.
4) **Large-scale benchmarking**: Evaluate on additional domains including cybersecurity (CICIDS), industrial IoT (SECOM), and the full ADBench [32] suite of 57 datasets.
5) **Adversarial robustness**: Test against adversaries who know the specific operators and attempt to evade all simultaneously.
6) **Semi-supervised SubspaceScan**: Extend the gated stretching mechanism to work without labels by using density-based discriminative scores instead of supervised Cohen's d.
7) **Multi-engine comparative study**: The UDP System Mode architecture supports alternative detection engines—Gravity (inter-agent gravitational coupling) and Molecular (kinetic energy and thermal equilibrium)—that exploit different physical analogies for systemic anomaly aggregation. A companion paper will present per-bank ablations, multi-engine fusion strategies, and cross-sector deployment of these variants.

IX. Conclusion

We have presented the Universal Deviation Principle, a multi-spectrum framework for anomaly detection that:

1) Formalises the Universal Deviation Principle through six theorems with explicit assumptions (A1–A3): if the operator family is deviation-separating (full-rank stacked Jacobian), then any displacement from the normal reference must produce a measurable response in at least one operator (Theorems 1–2), with finite-sample guarantees (Theorem 3), SDP-optimal fusion weights (Theorem 4), submodular operator selection (Theorem 5), and a sufficient condition for detection separation (Theorem 6).
2) Implements this principle through 14 composable spectrum projection operators across five mathematical families, with a Boundary-Centred Dimension Magnifier paired with QDA for supervised detection and Hybrid auto-selection for coverage.
3) Introduces **SubspaceScan** with adaptive gated stretching (Eqs. (35)–(37)) to address TYPE-1 BOUNDARY anomalies, achieving **mAUC** 0.910 across seven ODDS benchmarks—a $+0.9$ pp improvement over MetaFusion-CombA (0.901).
4) Achieves robust five-fold cross-validation (0.925 ± 0.020) on mammography—within 0.2 pp of LightGBM (0.927) despite using a closed-form QDA head—while using a single set of hyperparameters across the benchmark suite.
5) Validates operator diversity through correlation analysis: among 91 operator pairs, only Wavelet↔Phase (ρ=0.92) is redundant.
6) Introduces top-k **coverage** as a complementary metric, revealing failure modes where high AUC does not translate to actionable detection. Hybrid auto-selection achieves 91% mean coverage.
7) Provides the first head-to-head comparison of UDP against gradient-boosted trees (XGBoost, LightGBM) on five ODDS datasets, demonstrating that UDP matches LightGBM on Mammography $(0.925$ vs $0.927)$ with only a closed-form QDA head and no gradient optimisation, and identifying high-dimensional scaling as the primary open limitation.
8) Reports prospective case studies on **real-world sensitive sectors**: 25 G-SIB banks under a strictly prospective protocol, FRED macroeconomic indicators, and ERCOT grid-failure data. These studies illustrate how the framework behaves on structured financial and infrastructure monitoring tasks, complementing—rather than replacing—the benchmark evaluation.
9) Extends the framework to **multi-class classification** via per-class MDN decomposition (Section VI), achieving mean accuracy 0.936 across six standard benchmarks—competitive with LightGBM (0.934) and RF-200 (0.933)—thereby demonstrating dual-use capa-

bility of the multi-spectrum representation for both anomaly detection and classification.

The framework generalises BSDT's post-model correction into a model-independent detection paradigm. By measuring deviation across complementary operators and rescuing boundary anomalies through adaptive subspace aggregation, the Universal Deviation Principle reframes anomaly detection as a structured multi-operator analysis problem. The classification extension further validates the quality of the multi-spectrum representation: the same operators that detect anomalies also discriminate classes, using only $C\times(K+2)$ MDN features with no gradient optimisation. The benchmark results establish core empirical performance across both tasks, while the financial, macroeconomic, and energy case studies illustrate how the framework can be instantiated on multi-agent monitoring tasks.

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Appendix

All proofs are self-contained; no external dependencies beyond standard analysis are required.

A. Proof of Theorem 1 (Operator Sensitivity — Local)

We must show that the stacked map $\Phi = (\Phi_1, \ldots, \Phi_K)$ is locally injective at $\boldsymbol{\mu}$ and satisfies the quantitative bound (3).

Step 1 (Full-rank Jacobian). Because $\{\Phi_k\}$ is deviation-separating (Definition 1), $\bigcap_{k=1}^{K} \ker D\Phi_k(\boldsymbol{\mu}) = \{0\}$. Hence the stacked Jacobian $J(\boldsymbol{\mu}) = [D\Phi_1(\boldsymbol{\mu})^\top, \ldots, D\Phi_K(\boldsymbol{\mu})^\top]^\top \in \mathbb{R}^{D\times m}$ has $\ker J(\boldsymbol{\mu}) = \{0\}$, so $\sigma_{\min}(J(\boldsymbol{\mu})) > 0$.

Step 2 (Inverse Function Theorem). Since $J(\boldsymbol{\mu})$ has full column rank, the Inverse Function Theorem guarantees that Φ is locally injective in a neighbourhood $U \ni \boldsymbol{\mu}$.

Step 3 (Quantitative bound). Taylor-expand each Φ_k about $\boldsymbol{\mu}$:

$$\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu}) = D\Phi_k(\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu}) + O(\|\mathbf{x}-\boldsymbol{\mu}\|^2).$$

Stacking: $\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu}) = J(\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu}) + O(\|\mathbf{x}-\boldsymbol{\mu}\|^2)$. By the singular-value lower bound, $\|J(\boldsymbol{\mu})\mathbf{v}\| \geq \sigma_{\min}\|\mathbf{v}\|$ for all $\mathbf{v}$, so

$$\|\Phi(\mathbf{x}) - \Phi(\boldsymbol{\mu})\| \geq \sigma_{\min}\|\mathbf{x}-\boldsymbol{\mu}\| - O(\|\mathbf{x}-\boldsymbol{\mu}\|^2).$$

Because $\max_k \|a_k\| \geq \|a\|/\sqrt{K}$ for any partitioning into K blocks, we obtain (3). □

B. Proof of Theorem 2 (Operator Sensitivity — Global)

The strategy extends Theorem 1 uniformly over a compact set $\mathcal{K}$.

Step 1 (Uniform lower bound). By hypothesis, $\sigma_{\min}(J(\mathbf{x})) > 0$ for every $\mathbf{x} \in \mathcal{K}$. Since $\sigma_{\min}(J(\cdot))$ is continuous and $\mathcal{K}$ is compact, the Extreme Value Theorem gives $\underline{\sigma} := \min_{\mathbf{x}\in\mathcal{K}} \sigma_{\min}(J(\mathbf{x})) > 0$.

Step 2 (Second-order remainder). Each $\Phi_k \in C^2$, so the Taylor remainder satisfies $\|R_k(\mathbf{x}, \boldsymbol{\mu})\| \leq \frac{1}{2}\sup_{\mathbf{z}\in\mathcal{K}} \|D^2\Phi_k(\mathbf{z})\| \cdot \|\mathbf{x}-\boldsymbol{\mu}\|^2$. Stacking and defining $C = \frac{1}{2}\sup_{\mathbf{z}\in\mathcal{K}} \|D^2\Phi(\mathbf{z})\|$ (finite by continuity on a compact set), we obtain a uniform remainder $\leq C\|\mathbf{x}-\boldsymbol{\mu}\|^2$.

Step 3 (Combine). For any $\mathbf{x} \in \mathcal{A} \subset \mathcal{K}$: $\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| \geq \underline{\sigma}\|\mathbf{x}-\boldsymbol{\mu}\|/\sqrt{K} - C\|\mathbf{x}-\boldsymbol{\mu}\|^2$. Taking the infimum over $\mathbf{x} \in \mathcal{A}$ yields (4). □

C. Proof of Theorem 3 (Finite-Sample Guarantee)

Step 1 (Mean estimation error). Since $\hat{\boldsymbol{\mu}}_n$ is the sample mean of n i.i.d. sub-Gaussian draws with covariance Σ, a standard Hoeffding-type bound gives: for each coordinate j, $\Pr[|\hat{\mu}_{n,j} - \mu_j| > t] \leq 2\exp\bigl(-nt^2/(2\Sigma_{jj})\bigr)$. A union bound over all m coordinates and choice $t_j = \sqrt{2\Sigma_{jj}\ln(2m/\delta)/n}$ yields $\Pr\bigl[\|\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}\| > \sqrt{\mathrm{tr}(\Sigma)} \cdot \sqrt{2\ln(2m/\delta)/n}\bigr] \leq \delta$.

Step 2 (Lipschitz transfer). Each Φ_k is L_k-Lipschitz, so:

$$\|\Phi_k(\mathbf{x}) - \Phi_k(\hat{\boldsymbol{\mu}}_n)\| \geq \|\Phi_k(\mathbf{x}) - \Phi_k(\boldsymbol{\mu})\| - L_k\|\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}\|.$$

Step 3 (Combine with Theorem 1). Applying the local bound (Theorem 1 at the population mean) and the $L = \max_k L_k$ simplification:

$$\max_k \|\Phi_k(\mathbf{x}) - \Phi_k(\hat{\boldsymbol{\mu}}_n)\| \geq \frac{\underline{\sigma}}{\sqrt{K}}\|\mathbf{x}-\boldsymbol{\mu}\| - \frac{L\sqrt{\mathrm{tr}(\Sigma)} \cdot \sqrt{2\ln(2m/\delta)}}{\sqrt{n}}$$

with probability $\geq 1-\delta$, yielding (5). □

D. Proof of Theorem 4 (SDP Formulation)

Step 1 (Concavity). The minimum eigenvalue $\lambda_{\min}(A) = \min_{\|\mathbf{v}\|=1} \mathbf{v}^\top A\,\mathbf{v}$ is the infimum (equivalently, pointwise minimum) of linear functions of A. Since $\sum_k w_k\mathcal{I}_k$ is linear in $\mathbf{w}$, the composition $\eta(\mathbf{w}) = \lambda_{\min}(\sum_k w_k\mathcal{I}_k)$ is concave on Δ_K as an infimum of affine functions.

Step 2 (SDP reformulation). Maximising a concave function over a polytope is a convex programme. The specific structure—maximise $\eta(\mathbf{w})$ subject to $\mathbf{w} \in \Delta_K$—admits the standard SDP formulation:

$$\max_{\mathbf{w},\,\eta}\ \eta \quad \text{s.t.} \quad \textstyle\sum_k w_k\mathcal{I}_k \succeq \eta I, \quad \mathbf{w} \in \Delta_K,$$

which is solvable in polynomial time by interior-point methods. □

E. Proof of Theorem 5 (Greedy Operator Selection)

Step 1 (Monotonicity). Adding an operator increases $\sum_{k\in S}\mathcal{I}_k$ in the positive-semidefinite order, so $\log\det(\sum_{k\in S}\mathcal{I}_k)$ is non-decreasing.

Step 2 (Submodularity). For sets $S \subseteq T$ and a new operator $j \notin T$:

$$f(S\cup\{j\}) - f(S) = \log\det(I + (\textstyle\sum_{k\in S}\mathcal{I}_k)^{-1}\mathcal{I}_j).$$

As S grows, $(\sum_{k\in S}\mathcal{I}_k)^{-1}$ decreases (in the Loewner order), so the marginal gain diminishes. This is the defining property of submodularity.

Step 3 (Greedy guarantee). By the classical result of Nemhauser, Wolsey, and Fisher (1978), greedy maximisation of a monotone submodular function subject to a cardinality constraint $|S| \leq K$ achieves at least $(1-1/e) \approx 0.632$ of the optimal value, yielding (7). □

F. Proof of Theorem 6 (Operator-Space Separation)

Step 1 (Nearest-point projection). Since $\mathcal{N}$ is compact and $\mathbf{x} \notin \mathcal{N}$, there exists $\mathbf{z}^* = \arg\min_{\mathbf{z}\in\mathcal{N}} \|\mathbf{x} - \mathbf{z}\|$ with $d := \|\mathbf{x} - \mathbf{z}^*\| > 0$.

Step 2 (Local expansion at $\mathbf{z}^*$). Apply the Taylor expansion used in Theorem 1 at $\boldsymbol{\mu} = \mathbf{z}^*$:

$$\|\Phi(\mathbf{x}) - \Phi(\mathbf{z}^*)\| \geq \sigma_{\min}(J(\mathbf{z}^*)) \cdot d - O(d^2).$$

Since $\sigma_{\min}(J(\mathbf{z}^*)) \geq \underline{\sigma} > 0$ by hypothesis, this gives $\|\Phi(\mathbf{x}) - \Phi(\mathbf{z}^*)\| \geq \underline{\sigma}\, d - O(d^2)$.

Step 3 (Score bound). By definition, $s(\mathbf{x}) = \inf_{\mathbf{z}\in\mathcal{N}} \|\Phi(\mathbf{x}) - \Phi(\mathbf{z})\| \leq \|\Phi(\mathbf{x}) - \Phi(\mathbf{z}^*)\|$. Conversely, for any $\mathbf{z} \in \mathcal{N}$ with $\|\mathbf{x} - \mathbf{z}\| \geq d$, the same local analysis yields $\|\Phi(\mathbf{x}) - \Phi(\mathbf{z})\| \geq \underline{\sigma}\, d - O(d^2)$, so $s(\mathbf{x}) \geq \underline{\sigma}\, d - O(d^2)$, establishing (9).

The per-operator bound $s_k(\mathbf{x}) \geq \underline{\sigma}\, d/\sqrt{K} - O(d^2)$ follows from the block decomposition argument in Theorem 1 (Step 3). □

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